

Thermal modular commutators as joint probes of gauge flux and broken time-reversal symmetry in axial spin networks

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Abstract

We establish an exact high-temperature onset law for the thermal modular commutator on finite, simple, connected, unicyclic spin networks with axial XXZ–Dzyaloshinskii–Moriya interactions. The site spins are arbitrary irreducible representations, and the longitudinal exchanges, longitudinal fields, and trees grafted onto the cycle are unrestricted. If the unique cycle γ has length n and contains three distinct marked sites A, B, C in this cyclic order, then the leading admissible coefficient is $2(-1)^n h_B \prod_{x \in V(\gamma)} [s_x(s_x + 1)/3] \operatorname{Im} W_\gamma$ at order β^{n+1} . The marked sites need not be consecutive. The proof uses gauge-flow grading, exact spin moments, and primitive cumulants of the three marked arcs. Coefficient extraction also gives an ambient-independent primitive contribution for each simple cycle that contains A, B, C and an exact shortest-cycle sum at the first admissible order. Two independent mechanisms force all-temperature vanishing within the axial class, namely a local real structure and physical time-reversal symmetry at zero field. Axiality is essential. Transverse on-site fields generate a local, non-Wilson contribution at order seven on a consecutive three-qubit path, and an exact seven-cycle example develops this contribution before its winding term. We also derive the complete spin-1/2 triangle expansion through sixth order and, in the XY specialization, its first higher Wilson harmonic. An exact Jordan–Wigner reconstruction applies to axial XY rings. For rings with homogeneous hopping magnitudes and a fixed marked-site field, the computed modular size ratios through $n = 70$ approach an exactly known fermion-parity-string scale. This last identification is numerical, while the determinant formula and its temperature asymptotics are analytic.

1 Introduction

Kim *et al.* proposed the modular commutator as a bulk wave-function probe of chiral central charge [1] and subsequently developed its general state-level properties [2]. Zou *et al.* derived its conformal-field-theory dependence on chiral central charge and the conformal cross ratio [3]. Fan related entanglement-generated dynamics to the gravitational anomaly [4]. Park, Kim, and Moon proved geometric additivity for the multipartite configurations they consider in the invertible-state area-law setting [5]. Gass and Levin introduced a Rényi-like generalization [6]. Zang studied responses of entanglement measures under modular flow [7]. Zeng found a robust half-quantized modular commutator at the single-Dirac-node critical point of the Haldane model and related it to the parity anomaly [8]. Sheffer *et al.* used a finite number of replicas with permutation defects to extract chiral central charge and Hall conductance [9]. Maity *et al.* applied modular commutators to the Zeeman–Kitaev honeycomb model and the kagome antiferromagnet with scalar-spin-chirality perturbations [10].

The same literature delimits the scope. Gass and Levin constructed lattice systems with spurious modular commutators, disproving a universal relation to chiral central charge [11]. Sun *et al.* showed that familiar pure-state chirality diagnostics can fail in mixed-state topological phases [12]. These limitations motivate a narrower question: what can the modular commutator detect exactly in a finite thermal spin

network, without assigning it a quantized or topological meaning? Problems 3 and 7 of Ref. [2] ask, respectively, for modular-commutator calculations in further chiral systems, with free fermions given as one direction, and for a characterization of chirality in few-body quantum systems. The present work gives exact finite-temperature realizations relevant to both questions, first in interacting axial spin networks and then in axial XY rings through an exact free-fermion representation.

Within the axial class, we find that a nonzero response requires both a nonreal Wilson product and broken time-reversal symmetry. On a unicyclic network, the leading admissible flux term is a winding process whose order grows with the cycle length. In the homogeneous-hopping XY-ring family studied below, the finite-temperature signal is found numerically to approach the fermion-parity-string rate over the computed sizes.

This narrower role is naturally connected with recent state-level notions of chirality and imaginarity. Vardhan *et al.* define multipartite chirality by the absence of a product local unitary mapping a state to its complex conjugate. They introduce computable nested-commutator witnesses, including the ordinary tripartite modular commutator, which are not faithful in general [13]. Fernandes *et al.* construct unitary-invariant witnesses of quantum imaginarity for sets of states [14]. A distinct many-body notion uses finite-depth local operations and can remain nontrivial even when tripartite modular commutators vanish [15]. Our contribution is not a new definition of state chirality. It is an exact thermal resolution, within a specified interacting spin class, of when the ordinary tripartite modular commutator must vanish and which gauge-invariant couplings control its first admissible terms.

We answer this question for axial XXZ–Dzyaloshinskii–Moriya (DM) models. The antisymmetric exchange interaction was introduced phenomenologically by Dzyaloshinsky and derived microscopically from spin-orbit-induced superexchange by Moriya [16, 17]. For an axial DM vector, the transverse exchange takes the complex-amplitude form used below. A phase on one bond depends on the choice of local spin axes, whereas the product accumulated around a closed cycle is invariant under independent rotations about the axial direction. We call this continuous, Abelian obstruction an irreducible spin-orbit gauge flux, or a Wilson-loop-type phase. No quantization or topological protection is implied.

For a faithful state ρ and two regions X, Y , we use

$$\mathfrak{M}_\rho(X, Y) = i \operatorname{Tr} \rho [\widehat{\log \rho_X}, \widehat{\log \rho_Y}]. \quad (1)$$

Here a hat denotes extension by the identity to $X \cup Y$ after the complement has been traced out. This is the modular commutator defined in Eq. (2) of Ref. [1] and Eq. (4) of Ref. [2]. For overlapping X, Y , set $A = X \setminus Y$, $B = X \cap Y$, and $C = Y \setminus X$, after tracing out the complement of $X \cup Y$. The two minus signs in the convention $K_R = -\log \rho_R$ cancel. Thus Eq. (1) is not a new generalization of the state functional. It is real, antisymmetric under $X \leftrightarrow Y$, and depends only on $\rho_{X \cup Y}$.

Principal results. First, we turn the local real-structure criterion of Ref. [2] into a graph-theoretic statement for axial spin networks. Corollary 3 states that if all fundamental Wilson products are real, then (1) vanishes for every pair of regions and at every temperature. Every tree is included. Independently, Corollary 4 shows that physical time reversal forces exact vanishing whenever all magnetic fields are zero, even in the presence of a nonreal Wilson product. A nonzero signal therefore requires a joint obstruction, namely irreducible gauge flux and broken time-reversal symmetry, subject to the chosen tripartition and any remaining spatial symmetries.

Second, Theorem 5 gives the leading admissible coefficient on every finite, simple, connected, unicyclic axial network with arbitrary irreducible site spins, longitudinal couplings, longitudinal fields, and grafted trees. If the unique cycle γ has length n and is oriented through three distinct marked sites A, B, C in this cyclic order, then

$$\mathfrak{M}_\beta(AB, BC) = 2(-1)^n \beta^{n+1} h_B \left(\prod_{x \in V(\gamma)} \kappa_x \right) \operatorname{Im} W_\gamma + O(\beta^{n+2}), \quad \kappa_x = \frac{s_x(s_x + 1)}{3}. \quad (2)$$

The sites need not be consecutive. Here AB denotes the subsystem $\{A, B\}$, whether or not it is an edge. The proof replaces the effective hopping substitution used in the previous ring argument by a dimensionless cumulant of the three marked arcs. Proposition 12 then identifies the primitive coefficient carried by any simple cycle in a general graph, while Corollary 13 gives the exact sum over shortest marked cycles at the first admissible order. Neither statement is an all-orders decomposition into independent cycle responses.

Third, Section 5 shows that the axial hypothesis cannot be removed. Consecutive transverse on-site fields on a three-qubit path produce the ambient-independent projected sector

$$\Pi_{\text{loc}} m_7 = \frac{J_{AB}^z h_B^z |t_{BC}|^2}{23040} \text{Im}(t_{AB} \bar{g}_A g_B). \quad (3)$$

This is a local open-link gauge invariant, not a Wilson-loop contribution. The complete m_7 is also classified for one explicitly stated eight-parameter open-path submodel. An exact seven-cycle example has the nonzero coefficient $m_7 = 1/23040$, while its primitive winding term occurs at order eight. Two later sections record further controlled directions. For a spin-1/2 axial triangle, Theorems 16 and 17 determine the complete fifth- and sixth-order coefficients, and Theorem 18 identifies the first higher Wilson harmonic when $J^z = 0$. For axial XY rings, Proposition 21 gives an exact all-temperature Jordan–Wigner reduction to four Gaussian parity components and fixed reduced blocks. The exact parity determinant defines a scale r_P . In the homogeneous-hopping family with a fixed marked-site field, the computed modular size ratios through $n = 70$ approach this scale over the temperatures studied. The determinant identity and its asymptotics are analytic. The identification of the modular decay rate remains numerical.

For the oriented triangle $A \rightarrow B \rightarrow C \rightarrow A$, write $W_\Delta = t_{AB} t_{BC} t_{CA}$ and $\Phi_\Delta = \text{Im } W_\Delta$. Fourth, we compare three diagnostics analytically in the same normalization. The scalar spin chirality begins at order β^2 but responds to individual, gauge-removable DM amplitudes. A product of oriented two-point coherences begins at order β^3 and measures Φ_Δ without a field. The modular commutator begins at order β^4 and measures $h_B \Phi_\Delta$. As a direct flux meter, the coherence loop therefore has a lower-order onset and requires only three two-point coherences. The distinct point of (1) is that it is a functional of the state and the tripartition alone. No bond operators, spin axes, or cycle need be specified in its definition.

The paper is organized as follows. Section 2 reviews the state-level structure and derives the two exact vanishing mechanisms for axial networks. Section 3 proves the arbitrary-spin unicyclic theorem. Section 4 isolates primitive cycle sectors on general graphs, and Sec. 5 establishes the necessity of axiality. Section 6 gives the qubit triangle refinements. Section 7 develops the exact Jordan–Wigner reconstruction and the numerical attenuation study. The final sections compare three chirality diagnostics, discuss finite-temperature behavior and measurement, and state the limits of the results.

2 State structure and exact vanishing in axial networks

Let $\mathcal{H} = \bigotimes_{x \in V} \mathcal{H}_x$ be finite dimensional, and let $\rho > 0$ be a faithful state. For a region $X \subseteq V$, write $\rho_X = \text{Tr}_{X^c} \rho$. If O_X acts on $\mathcal{H}_X$, its extension by the identity to $\mathcal{H}$ is denoted by $\hat{O}_X$.

2.1 Local-unitary and antiunitary transformations

The transformation laws below use the site-by-site specialization of the region-by-region product unitaries surrounding Eqs. (8) and (9) of Ref. [2]. No invariance under a circuit crossing the region boundaries is used.

Lemma 1 (Transformation laws) *Let $X, Y \subseteq V$.*

If $U = \bigotimes_x U_x$ is a product of local unitaries, then

$$\mathfrak{M}_{U\rho U^\dagger}(X, Y) = \mathfrak{M}_\rho(X, Y). \quad (4)$$

If $\mathcal{A} = VK$ is an antiunitary compatible with the product structure, where V is a product of local unitaries and K is complex conjugation in a product basis, then

$$\mathfrak{M}_{\mathcal{A}\rho\mathcal{A}^{-1}}(X, Y) = -\mathfrak{M}_\rho(X, Y). \quad (5)$$

Proof. The reduced state of $U\rho U^\dagger$ on X is $U_X\rho_X U_X^\dagger$. Spectral functional calculus therefore gives $\log(U_X\rho_X U_X^\dagger) = U_X(\log \rho_X)U_X^\dagger$, and Eq. (4) follows from cyclicity of the trace.

It is enough to prove Eq. (5) for K . Put $Z = \text{Tr} \rho[\widehat{\log \rho_X}, \widehat{\log \rho_Y}]$. Since the commutator is anti-Hermitian, Z is purely imaginary. Complex conjugation commutes with partial traces and spectral functional calculus, whence

$$\mathfrak{M}_{\bar{\rho}}(X, Y) = i \bar{Z} = -iZ = -\mathfrak{M}_\rho(X, Y).$$

The local-unitary part V is handled by Eq. (4). $\square$

2.2 Vanishing from a local real structure

The following criterion is likewise a known state-level property of the modular commutator [2]. Its proof is included to make clear which part of the later graph-theoretic corollary is model independent.

Proposition 2 (Local real-structure criterion) *Suppose that local unitaries U_x and a product basis exist such that $U\rho U^\dagger$ is real in that basis, where $U = \bigotimes_x U_x$. Then*

$$\mathfrak{M}_\rho(X, Y) = 0 \quad (6)$$

for every pair of regions $X, Y \subseteq V$.

Proof. By Lemma 1, it is enough to consider a real state ρ . Because ρ is Hermitian, its matrix is then real symmetric. Partial traces in the chosen product basis preserve reality and symmetry, and so does spectral functional calculus. Consequently, $\widehat{\log \rho_X}$ and $\widehat{\log \rho_Y}$ are real symmetric, whereas their commutator is real antisymmetric. If S is symmetric and A is antisymmetric, then $\text{Tr}(SA) = \text{Tr}[(SA)^T] = -\text{Tr}(AS) = -\text{Tr}(SA)$. Taking $S = \rho$ proves Eq. (6). $\square$

Reference [2] also noted the resulting vanishing for reduced Gibbs states of stoquastic Hamiltonians. Corollary 3 below has a different, model-specific purpose: it gives a necessary and sufficient cycle criterion for when an axial XXZ–DM Hamiltonian can be made real by local $U(1)$ rotations, even though its bond amplitudes are complex in the original S^z basis.

2.3 Axial XXZ–DM networks and Wilson products

At each vertex x , let $\mathcal{H}_x$ carry the irreducible spin- s_x representation, where

$$s_x \in \left\{ \frac{1}{2}, 1, \frac{3}{2}, \dots \right\}, \quad \kappa_x = \frac{s_x(s_x + 1)}{3}. \quad (7)$$

Consider the axial Hamiltonian

$$H = \sum_{\{x,y\} \in E} \left[\frac{1}{2} (t_{xy} S_x^+ S_y^- + \overline{t_{xy}} S_x^- S_y^+) + J_{xy}^z S_x^z S_y^z \right] - \sum_{x \in V} h_x S_x^z. \quad (8)$$

Here $t_{xy} \in \mathbb{C}$, $J_{xy}^z = J_{yx}^z \in \mathbb{R}$, and $h_x \in \mathbb{R}$. An orientation is chosen for each edge, with

$$t_{yx} = \overline{t_{xy}}, \quad t_{xy} = J_{xy}^\perp + iD_{xy}^z. \quad (9)$$

For the local rotation

$$U_\varphi = \exp \left(i \sum_x \varphi_x S_x^z \right), \quad (10)$$

one has

$$U_\varphi H(t, J^z, h) U_\varphi^\dagger = H(t^\varphi, J^z, h), \quad t_{xy}^\varphi = t_{xy} e^{i(\varphi_x - \varphi_y)}. \quad (11)$$

For an oriented cycle $\gamma = (x_1, x_2, \dots, x_n, x_1)$, define

$$W_\gamma = \prod_{j=1}^n t_{x_j x_{j+1}}, \quad \vartheta_\gamma = \arg W_\gamma, \quad \Phi_\gamma = \text{Im } W_\gamma. \quad (12)$$

The product W_γ and, when nonzero, its phase are invariant under Eq. (11).

Corollary 3 (Axial real-gauge criterion) *Remove all edges for which $t_{xy} = 0$. A gauge of the form Eq. (10) that makes every remaining t_{xy} real exists if and only if*

$$W_\gamma \in \mathbb{R} \quad (13)$$

for every fundamental cycle associated with a spanning forest, equivalently for every simple oriented cycle of the remaining graph. If this condition holds, then the Gibbs state $\rho_\beta = e^{-\beta H} / \text{Tr } e^{-\beta H}$ satisfies

$$\mathfrak{M}_\beta(X, Y) = 0 \quad (14)$$

for all X, Y and all finite β . In particular, Eq. (14) holds on every tree.

Proof. If every transformed coupling is real, every cycle product is real. For the converse, choose a spanning forest. On each component, fix the phase at one root and determine the remaining vertex phases recursively so that the couplings on the forest are real. Each edge outside the forest closes a unique fundamental cycle. Its transformed phase is the phase accumulated around that cycle, which belongs to $\pi\mathbb{Z}$ by Eq. (13). Hence every remaining coupling is real.

In this gauge the matrix of H is real in the product S^z basis, as are $e^{-\beta H}$ and the normalized Gibbs state. Proposition 2 completes the proof. $\square$

Complex conjugation K in the product S^z basis and physical time reversal Θ are distinct antiunitary operations. Their parameter actions are

$$KH(t, J^z, h)K^{-1} = H(\bar{t}, J^z, h), \quad (15)$$

$$\Theta H(t, J^z, h)\Theta^{-1} = H(t, J^z, -h). \quad (16)$$

The DM interaction is bilinear in spin operators and is therefore even under physical time reversal. Lemma 1 gives

$$\mathfrak{M}_\beta(\bar{t}, J^z, h) = -\mathfrak{M}_\beta(t, J^z, h), \quad (17)$$

$$\mathfrak{M}_\beta(t, J^z, -h) = -\mathfrak{M}_\beta(t, J^z, h). \quad (18)$$

Equation (18) gives a second exact vanishing mechanism that does not follow from the real-gauge criterion.

Corollary 4 (Zero-field vanishing) *For the axial Hamiltonian (8), if $h_x = 0$ at every vertex, then*

$$\mathfrak{M}_\beta(X, Y) = 0 \quad (19)$$

for all regions X, Y and all finite β , irrespective of the Wilson products.

Proof. At zero field the Gibbs state is invariant under physical time reversal. Antiunitary oddness then identifies $\mathfrak{M}_\beta$ with its negative. $\square$

Corollaries 3 and 4 are independent. A nonreal Wilson product obstructs the axial real gauge but does not by itself produce a modular signal. Conversely, a magnetic field breaks physical time reversal but produces no signal on a tree. Even when both obstructions are present, the particular regions or a reflection symmetry may still force an additional cancellation.

3 Arbitrary-spin onset on finite unicyclic networks

We now determine the first coefficient allowed by gauge invariance and physical time reversal on a finite unicyclic graph. The result applies to arbitrary irreducible site spins. It also allows arbitrary longitudinal exchanges, longitudinal fields, and trees grafted onto the unique cycle. All Hamiltonian parameters are held fixed as $\beta \rightarrow 0$.

Let $G = (V, E)$ be finite, simple, connected, and unicyclic. Denote its unique cycle by γ and its length by $n \geq 3$. Choose three distinct vertices $A, B, C \in V(\gamma)$. Orient γ so that they occur in the cyclic order

$$\gamma = (A, \dots, B, \dots, C, \dots, A). \quad (20)$$

For this orientation, set

$$W_\gamma = \prod_{(x,y) \in \gamma} t_{xy}. \quad (21)$$

The notation AB below means the subsystem $\{A, B\}$. It does not assert that A and B are adjacent. The same convention applies to BC .

Theorem 5 (Arbitrary-spin unicyclic onset) *For the axial Hamiltonian (8), let every $\mathcal{H}_x$ carry the irreducible spin- s_x representation, with κ_x defined in Eq. (7). Assume $t_{xy} \in \mathbb{C}$ with $t_{yx} = \overline{t_{xy}}$, $J_{xy}^z \in \mathbb{R}$, and $h_x \in \mathbb{R}$. Then every Taylor coefficient of $\mathfrak{M}_\beta(AB, BC)$ through order n vanishes, and*

$$\mathfrak{M}_\beta(AB, BC) = 2(-1)^n \beta^{n+1} h_B \left(\prod_{x \in V(\gamma)} \kappa_x \right) \text{Im } W_\gamma + O(\beta^{n+2}). \quad (22)$$

No coupling is assumed to be nonzero. Equation (22) is therefore a statement about the leading admissible coefficient. Its displayed coefficient can vanish for a particular Hamiltonian.

The proof occupies the rest of this section. We first isolate the only possible monomial at order $n+1$. We then evaluate its coefficient by reducing the cycle to three primitive arc cumulants. This second step is needed because tracing out an arc generates multispin operators in addition to its endpoint hopping.

3.1 Gauge-flow grading and pruning

Write the analytic expansion near the maximally mixed state as

$$\mathfrak{M}_\beta(AB, BC) = \sum_{q \geq 0} \beta^q m_q. \quad (23)$$

Finite dimensionality makes this series convergent in a neighborhood of $\beta = 0$. The identity $\rho_\beta(H) = \rho_1(\beta H)$ shows that m_q is homogeneous of total degree q in the Hamiltonian parameters.

For coefficient extraction, first write $t_e = x_e + iy_e$. Each m_q is a real polynomial in x, y, J^z, h . Complexify this polynomial and make the invertible change of variables

$$z_e = x_e + iy_e, \quad w_e = x_e - iy_e. \quad (24)$$

The variables z_e and w_e are now algebraically independent. The physical coefficient is recovered on $w_e = \overline{z_e}$. Gauge invariance under the vertex torus is a polynomial identity, so distinct torus characters vanish coefficientwise. Equations (17) and (18) give the polynomial identities

$$m_q(w, z, J^z, h) = -m_q(z, w, J^z, h), \quad (25)$$

$$m_q(z, w, J^z, -h) = -m_q(z, w, J^z, h). \quad (26)$$

This formalization also makes every edge multidegree unambiguous.

Fix an orientation of each edge. For a monomial, let a_e and b_e be the exponents of z_e and w_e , and define the integral edge flow

$$d_e = a_e - b_e. \quad (27)$$

Local rotations about the z axis leave the modular commutator invariant. Character independence under this torus action means that the coefficient of a monomial can be nonzero only if

$$\text{div } d = 0 \quad (28)$$

at every vertex. On a unicyclic graph, any integral flow satisfying Eq. (28) vanishes on every bridge. Its restriction to γ is k times the consistently oriented unit flow for some $k \in \mathbb{Z}$. This follows by removing the cycle edges, starting at a leaf of each remaining tree, and imposing zero divergence inward. Zero divergence on the cycle then makes all oriented cycle values equal.

Let $D_{\text{hop}} = \sum_e (a_e + b_e)$ be the hopping degree. Equation (25) removes the sector $d = 0$. Indeed, every monomial in that sector is fixed by $z \leftrightarrow w$ after identical factors are collected. Every surviving monomial thus has $k \neq 0$, and

$$D_{\text{hop}} \geq \sum_e |d_e| = n|k| \geq n. \quad (29)$$

Equation (26) says that the total field degree of every surviving monomial is odd and in particular is at least one. In the physical Hamiltonian, this identity follows because time reversal leaves t and J^z fixed in the parameterization used here, sends h to $-h$, and reverses the modular commutator. Equations (23) and (29) now imply

$$m_q = 0 \quad (0 \leq q \leq n). \quad (30)$$

Equality in the degree bound at order $n + 1$ is rigid. The flow is one unit around γ , one oriented hopping monomial occurs on every cycle edge, and exactly one longitudinal field occurs. There is no

degree left for a longitudinal exchange, a bridge hopping, or a neutral return. After combining the two conjugate orientations, one obtains

$$m_{n+1} = \text{Im } W_\gamma \sum_{x \in V} c_x h_x \quad (31)$$

with real constants c_x that can a priori depend on the graph and its site spins.

Equation (31) also proves a useful pruning statement. Extract the coefficient of $h_x W_\gamma$ and set every parameter absent from that monomial to zero. Every grafted tree then becomes a decoupled tensor factor. If $x \notin V(\gamma)$, the remaining field acts on a factor disjoint from ABC , so the reduced state on ABC is independent of h_x and $c_x = 0$. If $x \in V(\gamma)$, every off-cycle normalized trace equals one. The coefficient is therefore exactly the coefficient of the isolated marked cycle, and the remaining constants depend only on the cycle spins. We may work with H_γ from now on.

3.2 Three-site expansion through fourth order

The triangular seed below uses a general local expansion that we record here. In this subsection, H is a Hamiltonian on $\mathcal{H}_A \otimes \mathcal{H}_B \otimes \mathcal{H}_C$. A scalar can be subtracted from H without changing its Gibbs state, so we take $\tau(H) = 0$, where

$$\tau(O) = \frac{1}{d} \text{Tr } O, \quad d = \dim(\mathcal{H}_A \otimes \mathcal{H}_B \otimes \mathcal{H}_C). \quad (32)$$

For $X \subseteq ABC$, let X^c denote the complement inside ABC and introduce the trace-preserving conditional expectation

$$\mathbb{E}_X(O) = \frac{1}{d_{X^c}} \text{Tr}_{X^c}(O) \otimes I_{X^c}. \quad (33)$$

Set

$$r_\beta = d\rho_\beta = \frac{e^{-\beta H}}{\tau(e^{-\beta H})}, \quad L_X(\beta) = \log \mathbb{E}_X(r_\beta). \quad (34)$$

Since $\mathbb{E}_X(r_\beta) = d_X \rho_{\beta,X} \otimes I_{X^c}$, L_X differs from the extended operator $\widehat{\log \rho_{\beta,X}}$ only by a scalar. Hence

$$\mathfrak{M}_\beta(AB, BC) = i\tau(r_\beta[L_{AB}(\beta), L_{BC}(\beta)]). \quad (35)$$

The normalized Gibbs density has the expansion

$$r_\beta = I - \beta H + \frac{\beta^2}{2} (H^2 - \tau(H^2)I) + O(\beta^3). \quad (36)$$

Define

$$P = \mathbb{E}_{AB}(H), \quad Q = \mathbb{E}_{BC}(H), \quad (37)$$

and, modulo scalar multiples of the identity,

$$A_2 = \frac{1}{2} (\mathbb{E}_{AB}(H^2) - P^2), \quad (38)$$

$$B_2 = \frac{1}{2} (\mathbb{E}_{BC}(H^2) - Q^2). \quad (39)$$

Expanding the logarithms gives

$$L_{AB}(\beta) = -\beta P + \beta^2 A_2 + O(\beta^3), \quad (40)$$

$$L_{BC}(\beta) = -\beta Q + \beta^2 B_2 + O(\beta^3). \quad (41)$$

Lemma 6 (Expansion through fourth order) *The coefficient of β^2 in the modular commutator vanishes. The next two coefficients are*

$$[\beta^3]\mathfrak{M}_\beta(AB, BC) = -i\tau(H[P, Q]), \quad (42)$$

$$[\beta^4]\mathfrak{M}_\beta(AB, BC) = i\tau \left[\frac{1}{2}H^2[P, Q] + H([P, B_2] + [A_2, Q]) \right]. \quad (43)$$

Proof. Insert Eqs. (36) through (41) into Eq. (35). At order β^2 one obtains $i\tau([P, Q]) = 0$. At order β^3 , every bare commutator has zero trace, leaving Eq. (42). At order β^4 , terms involving the third-order coefficients of L_{AB} or L_{BC} again occur as bare commutators. The remaining contributions are those in Eq. (43). $\square$

3.3 Local spin moments and the triangular seed

Let

$$\tau_x(O) = \frac{1}{2s_x + 1} \text{Tr}_x O. \quad (44)$$

Rotational invariance of the normalized trace and the spin commutation relations give

$$\tau_x(S_x^a) = 0, \quad (45)$$

$$\tau_x(S_x^a S_x^b) = \kappa_x \delta_{ab}, \quad (46)$$

$$\tau_x(S_x^a S_x^b S_x^c) = \frac{i\kappa_x}{2} \varepsilon_{abc}. \quad (47)$$

For completeness, the second moment is an invariant rank-two tensor. Its trace over $a = b$ is $s_x(s_x + 1)$, which gives Eq. (46). The invariant rank-three tensor is antisymmetric. Taking the trace of $[S_x^a, S_x^b]S_x^c = i\varepsilon_{abd}S_x^d S_x^c$ fixes its coefficient and proves Eq. (47). In the ladder basis these identities read

$$\tau_x(S_x^+ S_x^-) = \tau_x(S_x^- S_x^+) = 2\kappa_x, \quad (48)$$

while the six orderings of S_x^+, S_x^-, S_x^z have normalized trace $+\kappa_x$ or $-\kappa_x$, three of each.

Proposition 7 (Arbitrary-spin triangular seed) *For the isolated three-site axial triangle oriented as $A \rightarrow B \rightarrow C \rightarrow A$, with arbitrary irreducible spins and arbitrary axial couplings,*

$$\mathfrak{M}_\beta(AB, BC) = -2\beta^4 h_B \kappa_A \kappa_B \kappa_C \text{Im}(t_{AB} t_{BC} t_{CA}) + O(\beta^5). \quad (49)$$

The coefficient is independent of h_A , h_C , and all J^z couplings.

Proof. Homogeneity, gauge invariance, conjugation oddness, and field oddness give

$$[\beta^4]\mathfrak{M}_\beta(AB, BC) = (c_A h_A + c_B h_B + c_C h_C) \text{Im}(t_{AB} t_{BC} t_{CA}). \quad (50)$$

There is no degree available for J^z . Evaluate the three constants at the unit-amplitude parameter point

$$t_{AB} = t_{BC} = 1, \quad t_{CA} = i, \quad J^z = 0. \quad (51)$$

Use P, Q, A_2, B_2 from Eqs. (37) through (39) and write

$$T_0 = \frac{i}{2} \tau(H^2[P, Q]), \quad (52)$$

$$T_1 = i\tau(H[P, B_2]), \quad (53)$$

$$T_2 = i\tau(H[A_2, Q]). \quad (54)$$

Lemma 6 gives $[\beta^4]\mathfrak{M}_\beta = T_0 + T_1 + T_2$. Differentiation with respect to one field at zero field gives the exact contraction table

r	$\partial_{h_r} T_0$	$\partial_{h_r} T_1$	$\partial_{h_r} T_2$	$\partial_{h_r} m_4$
A	0	0	0	0
B	$2K_\Delta$	$-2K_\Delta$	$-2K_\Delta$	$-2K_\Delta$
C	0	0	0	0

$$K_\Delta = \kappa_A \kappa_B \kappa_C. \quad (55)$$

We justify the lift from the explicit spin-1/2 contraction in Appendix A. In every nonzero square-free contraction of $h_r t_{AB} t_{BC} t_{CA}$, each vertex distinct from r carries one S^+ and one S^- . The vertex r carries one each of S^+ , S^- , and S^z . If a conditional expectation splits the letters at a vertex among two normalized traces, one factor contains a single traceless generator and the contraction vanishes. Every surviving contraction therefore uses exactly one of Eqs. (48) and (47) at each vertex. Its ratio to the spin-1/2 value is $4\kappa_x$ at vertex x . Multiplying every entry of the qubit table by $\prod_{x \in \{A, B, C\}} 4\kappa_x$ gives Eq. (55). Summing its three columns proves Eq. (49). $\square$

This argument lifts the square-free contraction, not the Pauli anticommutator identities. Only the second and third normalized spin moments are used.

3.4 Primitive cumulants of the three marked arcs

Split the oriented cycle into the three internally disjoint arcs

$$P_1 : A \longrightarrow B, \quad P_2 : B \longrightarrow C, \quad P_3 : C \longrightarrow A. \quad (56)$$

Let their lengths be $\ell_1, \ell_2, \ell_3 \geq 1$, so that $\ell_1 + \ell_2 + \ell_3 = n$. Put $U = \{A, B, C\}$ and $I = V(\gamma) \setminus U$. With

$$\tau_I(O) = \frac{1}{d_I} \text{Tr}_I O, \quad d_I = \prod_{x \in I} (2s_x + 1), \quad (57)$$

the normalized partial trace over I defines

$$G(\beta) = \tau_I(e^{-\beta H_\gamma}), \quad (58)$$

$$Q(\beta) = -\log G(\beta). \quad (59)$$

The physical reduced state on U is $G/\text{Tr}_U G$. Define the real-analytic functional near $Q = 0$

$$\mathcal{F}(Q) = \mathfrak{M}_{e^{-Q}/\text{Tr}_U e^{-Q}}(AB, BC). \quad (60)$$

Then $\mathfrak{M}_\beta(AB, BC) = \mathcal{F}(Q(\beta))$. The dimensionless definition (59) is essential. The composed functional is $\mathcal{F}(Q)$, not the same functional evaluated on $-\beta^{-1} \log G$.

Consider an oriented arc

$$P = (v_0 = u, v_1, \dots, v_\ell = v), \quad T_P = \prod_{j=0}^{\ell-1} t_{v_j v_{j+1}}, \quad (61)$$

and let $P^\circ = \{v_1, \dots, v_{\ell-1}\}$. The symbol $[P]$ denotes extraction of the primitive multiaffine coefficient in which every oriented hopping of P occurs once and no other Hamiltonian parameter occurs. The monomial T_P is retained by this notation.

Lemma 8 (Primitive arc jet) *For every marked arc P ,*

$$[P]G = (-\beta)^\ell \left(\prod_{x \in P^\circ} \kappa_x \right) \frac{T_P}{2} S_u^+ S_v^-. \quad (62)$$

Consequently,

$$[P]Q = -[P]G. \quad (63)$$

Proof. Only the term of degree ℓ in the exponential can contribute to $[P]G$. The bond normalization gives $2^{-\ell}$. At every internal vertex the two possible local orders are $S^- S^+$ and $S^+ S^-$. Both have normalized trace $2\kappa_x$ by Eq. (48). All $\ell!$ global orderings therefore give the same partially traced operator, and they cancel the factorial in the exponential. The resulting scalar is $2^{-\ell} \prod_{x \in P^\circ} 2\kappa_x = \frac{1}{2} \prod_{x \in P^\circ} \kappa_x$, which proves Eq. (62).

To extract $[P] \log G$, expand the logarithm around $G = I$. Every nonlinear term would split the edge set of P into two or more nonempty proper subsets. Lemma 9, proved next, makes each such factor zero. Thus $[P] \log G = [P]G$, and the minus sign in Eq. (59) proves Eq. (63). $\square$

Lemma 9 (Proper-subpath vanishing) *In the primitive edge sector of the three marked arcs, a coefficient block means one factor arising in an ordered product from the exponential or logarithmic expansion. Such a coefficient of G vanishes if its selected edges contain a nonempty proper subset of at least one arc and do not contain the rest of that arc in the same block. The conclusion remains true when the block contains edges from the other arcs and the unique longitudinal field.*

Proof. The selected edges of a proper arc subset have a connected component with an endpoint in the interior of that arc. At this endpoint the local operator has axial charge $+1$ or -1 . Every other selected edge in the same block lies either on the same proper subset or on an arc with a disjoint interior, so it cannot cancel this local charge. A longitudinal field has charge zero. The normalized trace of a charged local operator vanishes under rotations about the z axis. Hence the entire partial trace is zero. $\square$

Lemma 10 (Pairwise shuffle cancellation) *Let P_i and P_j be two complete marked arcs. In the primitive sector with no further insertion,*

$$[P_i P_j]G = \frac{1}{2} ([P_i]G[P_j]G + [P_j]G[P_i]G). \quad (64)$$

The same identity holds for a complete arc and one field on a retained vertex. It follows that

$$[P_i P_j]Q = 0, \quad [P_i h_r]Q = 0 \quad (r \in U). \quad (65)$$

If the field is on an internal vertex of P_i , then

$$[P_j h_r]Q = 0 \quad (j = 1, 2, 3) \quad (66)$$

as well.

Proof. The raw coefficient of two complete arcs is the sum over all shuffles of their bond letters. The internal traces are independent of the order by Eq. (48). Exactly half of the shuffles put the endpoint letter of P_i before the endpoint letter of P_j , and half put it after. This proves Eq. (64). The argument for an arc and a retained field is identical.

By Lemma 9, the quadratic term is the only nonlinear term in $\log G$ that can contribute to either two-ingredient coefficient. It is exactly one half of the symmetrized product in Eq. (64). Explicitly,

$$[P_i P_j]Q = -[P_i P_j]G + \frac{1}{2}([P_i]G[P_j]G + [P_j]G[P_i]G) = 0. \quad (67)$$

The same formula with one arc replaced by the retained field proves the second identity in Eq. (65).

Now place the field at an internal vertex r of P_i . The raw P_i -field coefficient contains the complete symmetrization of S_r^+, S_r^-, S_r^z . Its normalized trace is zero because the six traces in Eq. (47) cancel in three opposite pairs. The field singleton has trace $\tau_r(S_r^z) = 0$, so there is no logarithmic product to restore the coefficient. If $j \neq i$, the P_j coefficient factors from the one-site field trace and vanishes for the same reason. This proves Eq. (66) for every marked arc. $\square$

3.5 Composition with the modular functional

We need two elementary derivatives of the state functional at the maximally mixed state. We complexify its real-analytic derivatives when they act on the formal non-Hermitian oriented arc jets.

Lemma 11 (First two derivatives) *The functional in Eq. (60) satisfies*

$$D\mathcal{F}(0) = 0, \quad D^2\mathcal{F}(0) = 0. \quad (68)$$

Proof. For a perturbation R , each reduced logarithm is a scalar plus the linear term $-\mathbb{E}_X(R)$ and terms of higher degree, where $\mathbb{E}_X$ is the normalized conditional expectation onto X . Their commutator is therefore at least quadratic, which gives $D\mathcal{F}(0) = 0$. At quadratic order, the state in the expectation value may be replaced by the maximally mixed state. For two perturbations R, S , polarization leaves a sum of normalized traces of commutators such as $i\tau_U([\mathbb{E}_{AB}(R), \mathbb{E}_{BC}(S)])$. Every such trace is zero. The linear correction to the Gibbs weight first multiplies a quadratic commutator at cubic order. Hence $D^2\mathcal{F}(0) = 0$. $\square$

We can now finish the proof of Theorem 5.

Proof. The grading argument already proves Eq. (30) and reduces the coefficient at order $n+1$ to one cycle orientation and one field. By pruning, it remains to evaluate the isolated cycle.

First suppose that the field lies at one of the retained vertices. Regard the individual edge variables and the field variable as the variables of the composition $\mathcal{F}(Q)$. The multiaffine chain rule for the primitive coefficient is a sum over labeled set partitions, with no additional factorial. Lemma 9 says that any nonzero block containing arc edges must contain one or more complete arcs. We may therefore organize the partitions by the four ingredients

$$\{P_1, P_2, P_3, h_r\}. \quad (69)$$

Four singleton blocks contribute the polarized fourth derivative of $\mathcal{F}$. A pair and two singletons would contribute a third derivative, but every possible pair jet of Q vanishes by Lemma 10. A triple and a singleton, or two pairs, can contribute only through $D^2\mathcal{F}(0)$. One block can contribute only through $D\mathcal{F}(0)$. Lemma 11 removes the last two cases. Only the four singleton arc and field jets survive in this primitive multiaffine coefficient.

If the field lies in the interior of an arc, its singleton jet vanishes under the partial trace. A block containing the field and a proper part of that arc vanishes by Lemma 9. The block containing the field

and any complete arc has zero pair jet by Eq. (66). Every larger block meets either $D^2\mathcal{F}(0)$ or $D\mathcal{F}(0)$. Thus the coefficient of $h_r W_\gamma$ in m_{n+1} vanishes for every internal-cycle vertex r . The pruning argument gives the same conclusion for every off-cycle vertex. Of the retained fields, Proposition 7 retains only h_B .

For an arc $P : u \rightarrow v$ of length ℓ , its singleton jet can be written

$$[P]Q = \frac{\beta}{2} t_{uv}^{\text{eff}} S_u^+ S_v^-, \quad t_{uv}^{\text{eff}} = (-\beta)^{\ell-1} \left(\prod_{x \in P^\circ} \kappa_x \right) T_P. \quad (70)$$

Denote the reverse-oriented arc sector by $\overline{P}$. On the physical parameter slice it supplies the adjoint singleton, and

$$[P]Q + [\overline{P}]Q = \frac{\beta}{2} \left(t_{uv}^{\text{eff}} S_u^+ S_v^- + \overline{t_{uv}^{\text{eff}}} S_u^- S_v^+ \right). \quad (71)$$

These are identities only for primitive singleton jets of the dimensionless operator Q . No effective-Hamiltonian replacement is being made. The field singleton is $-\beta h_r S_r^z$. Polynomial complexification allows the polarized triangular coefficient in Proposition 7 to be applied orientation by orientation. The conjugate pair then recombines into $\text{Im } W_\gamma$. The powers of β are

$$\beta^4 \prod_{i=1}^3 \beta^{\ell_i-1} = \beta^{n+1}, \quad (72)$$

and the effective link factors give the sign

$$(-1)^{(\ell_1-1)+(\ell_2-1)+(\ell_3-1)} = (-1)^{n-3}. \quad (73)$$

The three arc products satisfy $T_{P_1} T_{P_2} T_{P_3} = W_\gamma$. Their internal spin factors, together with $\kappa_A \kappa_B \kappa_C$ from the triangular seed, give $\prod_{x \in V(\gamma)} \kappa_x$. Finally,

$$-2(-1)^{n-3} = 2(-1)^n. \quad (74)$$

This proves the coefficient in Eq. (22). Analyticity supplies the stated remainder for fixed Hamiltonian parameters. The proof was polynomial after formal complexification, so it also covers vanishing hoppings and hence $W_\gamma = 0$. $\square$

3.6 Orientation and nonconsecutive marked sites

No step above assumes that the marked vertices are consecutive. The three arc lengths enter the displayed leading coefficient only through their sum, the cycle product, and the local κ_x factors. Thus the coefficient in Eq. (22) is independent of the distribution of the three marked-site separations.

If an external orientation of γ is fixed instead of the convention in Eq. (20), define

$$\varepsilon_\gamma(A, B, C) = \begin{cases} +1 & \text{if } A, B, C \text{ occur in that cyclic order,} \\ -1 & \text{if } A, C, B \text{ occur in that cyclic order.} \end{cases} \quad (75)$$

The right-hand side of Eq. (22) then acquires the factor $\varepsilon_\gamma(A, B, C)$. Equivalently, one may always orient the cycle through A, B, C as in Eq. (20). Exchanging A and C reverses the result, in agreement with the antisymmetry of the modular commutator.

When $s_x = 1/2$ at every cycle vertex, $\kappa_x = 1/4$ and

$$2(-1)^n \prod_{x \in V(\gamma)} \kappa_x = \frac{(-1)^n}{2^{2n-1}}. \quad (76)$$

This is the qubit cycle-length law. The sharper even- n remainder available for an isolated qubit XY ring is a separate restricted statement. It is not part of Theorem 5.

4 Primitive cycle sectors on general axial graphs

The unicyclic theorem has a coefficient-level extension to an arbitrary finite axial graph. This extension concerns one specified coupling multidegree. It is not an expansion of the full answer as a sum of independent cycle functions.

Choose one reference orientation for every edge and formally complexify the hopping parameters by treating t_{xy} and $\bar{t}_{xy}$ as algebraically independent for each reference-oriented edge (x, y) . Define $t_{yx} = \bar{t}_{xy}$ and $\bar{t}_{yx} = t_{xy}$. The physical specialization is $\bar{t}_{xy} = \overline{t_{xy}}$. For an oriented simple cycle $\eta = (x_1, \dots, x_r, x_1)$, let

$$W_\eta = \prod_{j=1}^r t_{x_j x_{j+1}}, \quad \bar{W}_\eta = \prod_{j=1}^r t_{x_{j+1} x_j}, \quad x_{r+1} = x_1. \quad (77)$$

The second monomial is the reverse orientation of the first. Let Π_η project a polynomial onto the direct sum of the monomial spaces spanned by

$$\{h_x W_\eta, h_x \bar{W}_\eta : x \in V\}.$$

We define $\text{Prim}_\eta m_{r+1} = \Pi_\eta m_{r+1}$. Thus this sector contains no additional hopping, longitudinal exchange, or field factor. Finite dimensionality and the scaling $H \mapsto \lambda H$ imply that every m_q is a homogeneous polynomial of total parameter degree q .

Proposition 12 (Primitive cycle coefficient) *Let G be a finite simple graph. Equip it with the axial Hamiltonian (8) and arbitrary irreducible site spins. Let A, B, C be three distinct marked vertices and write*

$$\mathfrak{M}_\beta(AB, BC) = \sum_{q \geq 0} \beta^q m_q. \quad (78)$$

For every simple cycle η of length r ,

$$\text{Prim}_\eta m_{r+1} = 0 \quad (79)$$

unless $A, B, C \in V(\eta)$. If the cycle contains the three marked vertices, orient it so that they occur in the cyclic order $A \rightarrow B \rightarrow C \rightarrow A$. Then the formally complexified coefficient is

$$\boxed{\text{Prim}_\eta m_{r+1} = \frac{(-1)^r h_B}{i} \left(\prod_{x \in V(\eta)} \kappa_x \right) (W_\eta - \bar{W}_\eta).} \quad (80)$$

On the physical parameter slice,

$$\boxed{\text{Prim}_\eta m_{r+1} = 2(-1)^r h_B \left(\prod_{x \in V(\eta)} \kappa_x \right) \text{Im } W_\eta.} \quad (81)$$

No edge-disjointness assumption is imposed on the other cycles of G .

Proof. Fix $x \in V$ and extract the coefficients of $h_x W_\eta$ and $h_x \bar{W}_\eta$. Every parameter absent from these monomials is thereby set to zero. This includes all J^z couplings, all fields except h_x , and all hoppings outside η . Retain both formal hopping orientations on η . The surviving Hamiltonian is a tensor sum of the isolated cycle Hamiltonian and, when x lies outside the cycle, a one-site spectator Hamiltonian. Its Gibbs operator factorizes. Normalized partial traces remove every spectator factor exactly.

If η contains A, B, C , the arbitrary-spin unicyclic theorem applies to the physical slice of the isolated marked cycle. Since that identity holds for every complex hopping value, it extends polynomially to

independent t and $\bar{t}$. Projection onto $h_x W_\eta$ and $h_x \bar{W}_\eta$, for each x and both orientations, gives Eq. (80) and selects h_B . A field on a spectator component cannot enter the reduced state on ABC .

Suppose instead that η omits at least one marked vertex. If it omits C , coefficient extraction gives $\rho_{ABC} = \rho_{AB} \otimes \rho_C$. The modular commutator reduces to

$$\mathrm{i} \mathrm{Tr} \rho_{AB} [\log \rho_{AB}, I_A \otimes \log \rho_B] = 0.$$

The case in which A is omitted is analogous. If B is omitted, the lifted reduced logarithms are sums of one-site logarithms and commute. These three cases also cover cycles that omit two or all three marked vertices. Polynomial coefficient extraction therefore proves Eq. (79) on the original graph. $\square$

The proposition determines the complete first admissible coefficient when the shortest hopping cycles are considered together. Let G_t be the graph obtained by deleting the edges whose hopping amplitude is zero.

Corollary 13 (Girth coefficient) *Assume that G_t contains a cycle and let g be its girth. Then*

$$m_q = 0 \quad (0 \leq q \leq g) \quad (82)$$

and

$$m_{g+1} = 2(-1)^g h_B \sum_{\substack{\eta \subseteq G_t \text{ simple, } |\eta|=g \\ A,B,C \in V(\eta)}} \left(\prod_{x \in V(\eta)} \kappa_x \right) \mathrm{Im} W_\eta. \quad (83)$$

Each unoriented cycle is counted once and is oriented in the marked cyclic order. Equation (83) remains valid when distinct shortest cycles share edges or paths.

Proof. For any monomial in m_q , let a_e and b_e be the exponents of t_e and $\bar{t}_e$, and put $d_e = a_e - b_e$. Local $U(1)$ invariance makes d a divergence-free integral flow. Conjugation oddness excludes the zero flow. Every nonzero integral circulation decomposes into oriented simple cycles. Consequently,

$$\mathrm{deg}_{\mathrm{hop}} = \sum_e (a_e + b_e) \geq \sum_e |a_e - b_e| = \|d\|_1 \geq g.$$

Equality requires one unit of circulation around one simple g -cycle. Physical time-reversal oddness requires an odd field degree and therefore adds at least one further factor. This proves Eq. (82). At total degree $g+1$, no neutral return or J^z factor is possible. The coefficient is consequently the sum of the primitive sectors in Proposition 12, which gives Eq. (83). $\square$

If G_t is a forest, the local real-gauge criterion instead gives $\mathfrak{M}_\beta(AB, BC) = 0$ at every temperature. Corollary 13 is only a first-admissible-order statement. It does not establish a cactus formula and it does not imply cycle additivity at later orders.

Two exact higher-order calculations delimit this point. First, take five spin-1/2 sites and two triangles ABC and BDE that share only B . Set every J^z to zero, set $h_B = 1$ and all other fields to zero, and choose unit-modulus hoppings with

$$W_1 = W_{ABC} = \mathrm{i}, \quad W_2 = W_{BDE} = 1. \quad (84)$$

Exact coefficient extraction gives

$$m_7 = -\frac{1}{768}. \quad (85)$$

At hopping degree six, gauge flow restricts the mixed two-cycle sector on the unit-modulus phase torus to a linear combination of $\mathrm{Im}(W_1 W_2)$ and $\mathrm{Im}(W_1 \bar{W}_2)$. The graph automorphism $D \leftrightarrow E$ reverses the orientation of the second triangle while fixing A, B, C , and therefore equates the two coefficients. At

$W_1 = i$ and $W_2 = 1$, the single-cycle second harmonics vanish. Equation (85) therefore fixes the mixed sector as

$$-\frac{1}{1536} [\text{Im}(W_1 W_2) + \text{Im}(W_1 \overline{W_2})] = -\frac{1}{768} \text{Im } W_1 \text{Re } W_2. \quad (86)$$

The primary certificate also verifies the sign reversal under $W_2 \mapsto -W_2$. The rational value in Eq. (85) is reproduced by a second exact implementation of the same formal-series method. This is an implementation check, not an independent method.

Second, take the spin-1/2 triangle $A-B-D-A$ and attach the marked site C as a leaf at B . Set $J^z = 0$, $h_B = 1$, and all other fields to zero. Choose

$$t_{AB} = t_{BD} = t_{BC} = 1, \quad t_{DA} = i. \quad (87)$$

The only cycle omits C . The same primary exact certificate nevertheless gives

$$m_0 = m_1 = \dots = m_7 = 0, \quad m_8 = -\frac{7}{368640}. \quad (88)$$

Neutral returns on the leaf make the off-target cycle visible at this later order. Equations (86) and (88) show that the natural coefficient-level organization is by coupling multidegree, not by a sum of independent functions of Wilson products.

5 Why the axial hypothesis is necessary

The onset theorems above require the absence of on-site transverse fields. In this section we write $h_x^z \equiv h_x$ for the longitudinal field in Eq. (8). To locate the boundary of that hypothesis, add

$$H_\perp = -\frac{1}{2} \sum_x (g_x S_x^+ + \overline{g_x} S_x^-). \quad (89)$$

With $S^+ = S^x + iS^y$ and $H_\perp = -\sum_x (h_x^x S_x^x + h_x^y S_x^y)$, our convention is $g_x = h_x^x - ih_x^y$. Under the local rotation Eq. (10),

$$t_{xy} \mapsto e^{i(\varphi_x - \varphi_y)} t_{xy}, \quad g_x \mapsto e^{i\varphi_x} g_x. \quad (90)$$

The open-link phase $t_{AB} \overline{g_A} g_B$ is therefore gauge invariant. It is not a Wilson product and it need not involve any closed path. For a formally complexified monomial, let a_{xy} and b_{xy} be the exponents of t_{xy} and $\bar{t}_{xy}$ on each reference-oriented edge and set $d_{xy} = a_{xy} - b_{xy}$. Let $q_g(x)$ be the net power of g_x minus the net power of $\bar{g}_x$. Use

$$(\text{div } d)_x = \sum_{x \rightarrow y} d_{xy} - \sum_{y \rightarrow x} d_{yx}.$$

Gauge invariance now imposes

$$\text{div } d + q_g = 0. \quad (91)$$

Transverse insertions can therefore terminate an open hopping flow. The closed-circulation argument behind the girth bound no longer applies.

We first state the part of the order-seven coefficient that is independent of the ambient graph. The statement is a projected coefficient identity, not a classification of the full m_7 .

Proposition 14 (Ambient-independent transverse-field sector) *Let A, B, C be distinct vertices of a finite graph, with $\{A, B\}, \{B, C\} \in E$, and let these three vertices carry spin 1/2. Other vertices and couplings are arbitrary. Formally regard $t, \bar{t}, g, \bar{g}$ as independent variables and let Π_{loc} project m_7 onto the two multidegrees*

$$\begin{aligned} J_{AB}^z h_B^z t_{BC} \bar{t}_{BC} t_{AB} \bar{g}_A g_B, \\ J_{AB}^z h_B^z t_{BC} \bar{t}_{BC} \bar{t}_{AB} g_A \bar{g}_B. \end{aligned} \quad (92)$$

Then

$$\Pi_{\text{loc}} m_7 = \frac{J_{AB}^z h_B^z t_{BC} \bar{t}_{BC}}{46080i} (t_{AB} \bar{g}_A g_B - \bar{t}_{AB} g_A \bar{g}_B). \quad (93)$$

On the physical parameter slice this becomes

$$\boxed{\Pi_{\text{loc}} m_7 = \frac{J_{AB}^z h_B^z |t_{BC}|^2}{23040} \text{Im}(t_{AB} \bar{g}_A g_B)}. \quad (94)$$

Proof. Every m_q is a polynomial in the Hamiltonian parameters. Extracting either multidegree in Eq. (92) sets every absent parameter to zero. The resulting Hamiltonian is $H_{ABC} \otimes I_{\text{spectators}}$. The Gibbs operator factorizes, and the spectator dimension cancels in the partition function and in every normalized partial trace. The projected coefficient is therefore identical to the coefficient on the isolated open path $A-B-C$.

The remaining three-qubit coefficient is evaluated by the self-contained exact symbolic extraction described in Appendix E, over

$$\mathbb{Q}(i)[x_1, \dots, x_7] / (x_1^2, \dots, x_7^2).$$

It expands the normalized Gibbs state and both reduced logarithms through order seven before taking the modular commutator. The two coefficients are

$$\begin{aligned} [J_{AB}^z h_B^z t_{BC} \bar{t}_{BC} t_{AB} \bar{g}_A g_B] m_7 &= -\frac{i}{46080}, \\ [J_{AB}^z h_B^z t_{BC} \bar{t}_{BC} \bar{t}_{AB} g_A \bar{g}_B] m_7 &= \frac{i}{46080}. \end{aligned} \quad (95)$$

Discarding a square in the quotient ring is exact for these squarefree targets. Perturbative parameter degrees are nonnegative, so a term already divisible by x_j^2 cannot later contribute to either target monomial. Equation (93) follows, and physical specialization gives Eq. (94). $\square$

The projection in Proposition 14 is the ambient-independent statement. The complete coefficient can contain other local and nonlocal multidegrees. A full formula is available in one sharply delimited local model.

Proposition 15 (Complete isolated-path coefficient) *Consider exactly three qubits on the open path $A-B-C$. Set $h_A^z = h_C^z = 0$ and retain only the eight parameter families*

$$t_{AB}, t_{BC}, g_A, g_B, g_C, J_{AB}^z, J_{BC}^z, h_B^z. \quad (96)$$

Define the three gauge-invariant imaginary parts

$$\begin{aligned} I_{AB} &= \text{Im}(t_{AB} \bar{g}_A g_B), \\ I_{BC} &= \text{Im}(t_{BC} \bar{g}_B g_C), \\ I_{AC} &= \text{Im}(t_{AB} t_{BC} \bar{g}_A g_C). \end{aligned} \quad (97)$$

Then

$$m_0 = m_1 = \dots = m_6 = 0 \quad (98)$$

and

$$\begin{aligned} m_7 &= \frac{h_B^z}{23040} \left[J_{AB}^z (|t_{BC}|^2 - (J_{BC}^z)^2) I_{AB} \right. \\ &\quad + J_{BC}^z (|t_{AB}|^2 - (J_{AB}^z)^2) I_{BC} \\ &\quad \left. + \frac{J_{AB}^z J_{BC}^z}{2} I_{AC} \right]. \end{aligned} \quad (99)$$

Proof. The computer-assisted exact proof uses the self-contained extraction described in Appendix E. Its exhaustive polynomial expansion of the Gibbs and reduced-logarithm series produces ten formal complex monomials, paired into the five real sectors displayed in Eq. (99), and no other monomial through degree seven. The separate exact-arithmetic engine checks the physical formula at five specified nondegenerate Gaussian-rational points. It tests an independent implementation of the same formal-series method, not an independent method. Exhaustive monomial extraction supplies the coefficient identity. $\square$

If $J_{BC}^z = 0$, Eq. (99) reduces to Eq. (94) and g_C drops out. Formula (99) is complete only under Eq. (96). Enabling h_A^z , h_C^z , or additional ambient couplings permits further order-seven sectors.

Two finite rings show why this distinction matters for the axial onset law. Take consecutive marked sites $A = 0$, $B = 1$, and $C = 2$ on an oriented spin-1/2 ring. Choose

$$t_{AB} = 1 + i, \quad t_e = 1 \quad (e \neq AB), \quad J_{AB}^z = h_B^z = g_A = g_B = 1, \quad (100)$$

with every other J^z , longitudinal field, and transverse field equal to zero. On the six-cycle, the axial value obtained by setting $g_A = g_B = 0$ is

$$m_7^{\text{axial}} = \frac{1}{2048}. \quad (101)$$

At the point in Eq. (100), exact extraction from the unprojected series through order seven instead gives

$$m_7 = \frac{49}{92160} = \frac{1}{2048} + \frac{1}{23040}. \quad (102)$$

The local open-link sector and the primitive six-cycle sector occur at the same order. The exact values here and on the seven-cycle below are certified by the separate exact implementation summarized in Appendix E.

On the seven-cycle at the same parameter point, the same exact extraction gives

$$m_0 = m_1 = \dots = m_6 = 0, \quad m_7 = \frac{1}{23040}. \quad (103)$$

The primitive cycle multidegree is first admissible at order eight. Thus the full modular commutator is already nonzero before the winding contribution can occur. More generally, for a consecutive marked path on a spin-1/2 cycle of length $n \geq 7$, the certified local multidegree in Eq. (94) has fixed order seven, while the primitive cycle term has order $n + 1$. The separation between the two orders therefore grows with n .

These examples do not show that the order-seven term is the first transverse-field contribution in every model. They do not classify the complete transverse-field expansion, exclude lower-order sectors under other parameter choices, or give an order table for cycles of lengths three, four, and five. The local phase in Eq. (94) is loop independent, but it is not phase blind. It measures the phase of t_{AB} relative to the two on-site transverse-field directions.

No arbitrary-spin closed formula for the transverse-field background is asserted here.

One coefficient statement remains exact after transverse fields are added. The primitive $h_B W_\gamma$ coefficient of the axial theorem is unchanged, because extracting that multidegree sets every g_x and $\bar{g}_x$ to zero. It need not be the first term or the dominant term of the full modular commutator. Equations (102) and (103) prove that axially is essential to the leading-order theorem stated above.

6 Qubit triangle refinements

We now return to spin 1/2. The leading term is the $n = 3$ specialization of Theorem 5. The next coefficients resolve additional local couplings and higher harmonics of the same Wilson product.

6.1 Fifth order: the earliest endpoint channel

The same graph grading gives a short classification at the next order. Let D be the total hopping degree of a gauge-invariant monomial and let $d_e = n_e - m_e$ be its integer flow on each oriented edge. On a triangle, zero divergence implies $d = k(1, 1, 1)$. Moreover,

$$D \equiv \sum_e d_e = 3k \equiv k \pmod{2}. \quad (104)$$

At total degree five, antiunitary oddness and gauge invariance therefore leave only one field, one longitudinal exchange, and an oriented hopping triangle. Consequently

$$m_5 = \Phi_\Delta \sum_{x \in \{A, B, C\}} \sum_{e \in E} c_{x,e} h_x J_e^z. \quad (105)$$

The nine constants are fixed by taking one longitudinal edge at a time at the parameter point (51).

Conjugation oddness only requires $k \neq 0$. It does not require k to be odd. Flows $|k| \geq 2$ need hopping degree at least six and hence cannot occur in m_4 or m_5 . At order six, the candidate $k = \pm 2$ sector uses all six degrees in W_Δ^2 and leaves no field insertion, so it is excluded by the oddness under $h \mapsto -h$. It first becomes admissible at order seven as $h_x \text{Im}(W_\Delta^2)$.

Theorem 16 (Complete fifth-order coefficient) *For arbitrary axial couplings on the spin-1/2 triangle,*

$$m_5 = \frac{\Phi_\Delta}{384} [(3h_A - h_B - 2h_C)J_{AB}^z + (-2h_A - h_B + 3h_C)J_{BC}^z - 3h_B J_{CA}^z]. \quad (106)$$

Thus Eqs. (49) and (106) give the expansion through order β^5 for all axial couplings.

Proof. The symmetry classification gives Eq. (105). Exact evaluation of its nine numerical constants is summarized in Appendix B. Inserting that coefficient table yields Eq. (106). $\square$

Two consequences will be useful. If all longitudinal exchanges vanish, then $m_5 = 0$ identically for every hopping coupling. If the field is uniform, $h_A = h_B = h_C = h$, the two longitudinal edges incident on the overlap cancel and

$$m_5 = -\frac{h J_{CA}^z \Phi_\Delta}{128}. \quad (107)$$

The edge opposite the overlap is then the sole fifth-order channel.

6.2 Sixth and seventh orders

At total degree six, the surviving $k = \pm 1$ monomials are Φ_Δ times a cubic odd polynomial in the fields, $\Phi_\Delta h_x |t_e|^2$, and $\Phi_\Delta h_x J_e^z J_f^z$. Define the hopping contribution

$$\begin{aligned} \mathcal{P}_\perp = & 4h_B(h_A^2 + h_A h_C + h_B^2 + h_C^2) \\ & + h_A |t_{AB}|^2 + h_C |t_{BC}|^2 \\ & + 4h_B(|t_{AB}|^2 + |t_{BC}|^2) + 6h_B |t_{CA}|^2 \end{aligned} \quad (108)$$

and the longitudinal contribution

$$\begin{aligned} \mathcal{P}_z = & h_A [(J_{AB}^z)^2 + 3J_{AB}^z J_{CA}^z - 4J_{BC}^z J_{CA}^z] \\ & + h_B [(J_{AB}^z)^2 + (J_{BC}^z)^2 + 2(J_{CA}^z)^2 + J_{AB}^z J_{BC}^z - J_{BC}^z J_{CA}^z - J_{AB}^z J_{CA}^z] \\ & + h_C [(J_{BC}^z)^2 + 3J_{BC}^z J_{CA}^z - 4J_{AB}^z J_{CA}^z]. \end{aligned} \quad (109)$$

Theorem 17 (Complete sextic coefficient) *For arbitrary axial couplings on the spin-1/2 triangle,*

$$m_6 = \frac{\Phi_\Delta}{1536} (\mathcal{P}_\perp + \mathcal{P}_z). \quad (110)$$

Together with Proposition 7 and Theorem 16, this determines $\mathfrak{M}_\beta(AB, BC)$ through order β^6 .

Proof. The gauge-flow classification gives the three sectors above. The cubic field polynomial and the nine coefficients of $h_x |t_e|^2$ follow from an invertible exact calibration. The eighteen coefficients of $h_x J_e^z J_f^z$ are fixed by six exact longitudinal configurations with symbolic fields. Appendix B gives the coefficient tables and independent exact checks, yielding Eqs. (108) and (109). $\square$

Theorem 17 separates two routes to the endpoint fields. Longitudinal exchange is the earliest route, through m_5 . It is not the only one: at sixth order, h_A and h_C reappear without longitudinal exchange, weighted respectively by the squared amplitudes of the bonds AB and BC that join them to the overlap. In particular, in the XY triangle with $h_B = 0$,

$$\mathfrak{M}_\beta(AB, BC) = \frac{\beta^6 \Phi_\Delta}{1536} (h_A |t_{AB}|^2 + h_C |t_{BC}|^2) + O(\beta^7). \quad (111)$$

This last statement is not valid with arbitrary J^z , for which the quintic coefficient (106) can survive when $h_B = 0$.

The first higher-flow sector occurs one order later.

Theorem 18 (First higher Wilson harmonic) *Assume $J_e^z = 0$ on every edge. The seventh-order coefficient of the XY triangle is*

$$m_7 = -\frac{2h_A + 41h_B + 2h_C}{92160} \text{Im}(W_\Delta^2). \quad (112)$$

Proof. At total degree seven, gauge invariance, conjugation oddness, and time-reversal oddness leave only $h_x \text{Im}(W_\Delta^2)$. The three numerical constants are fixed by one exact evaluation over $\mathbb{Q}(i)$ with symbolic fields. A second generic Wilson product and a point with $\text{Im}(W_\Delta^2) = 0$ give independent exact checks. Details are collected in Appendix B. $\square$

Since $\text{Im}(W_\Delta^2) = 2 \text{Re } W_\Delta \text{Im } W_\Delta$, Eq. (112) is the first term sensitive to the real part of the Wilson product. It does not replace the leading flux response. It shows that the full thermal series resolves higher harmonics of the same gauge-invariant loop variable.

7 Exact finite-temperature reconstruction for XY rings

We now specialize to a spin-1/2 ring with $J^z = 0$. This specialization is used for an exact Jordan–Wigner reconstruction at arbitrary temperature. It also provides a nonperturbative check of the qubit ring specialization of Theorem 5. Label the sites by $0 = A$, $1 = B$, $2 = C$, and $3, \dots, n-1$. We set

$$H_n = \frac{1}{2} \sum_{j=0}^{n-1} \left(t_j S_j^+ S_{j+1}^- + \bar{t}_j S_j^- S_{j+1}^+ \right) - \sum_{j=0}^{n-1} h_j S_j^z, \quad (113)$$

$$W_n = \prod_{j=0}^{n-1} t_j. \quad (114)$$

Site indices in Eq. (113) are understood modulo n .

Corollary 19 (Leading response of a qubit XY ring) *For every $n \geq 3$ in the spin-1/2 XY ring (113),*

$$\mathfrak{M}_\beta(AB, BC) = (-1)^n \frac{\beta^{n+1} h_B}{2^{2n-1}} \text{Im } W_n + O(\beta^{n+2}). \quad (115)$$

If n is even, the remainder improves to $O(\beta^{n+3})$.

Proof. Since $\kappa_x = 1/4$ at every site, the leading formula follows directly from Theorem 5. No statement about the $J^z = 0$ specialization is used in that proof. For even n , the on-site transformation given in Appendix D implies $\mathfrak{M}_{-\beta} = -\mathfrak{M}_\beta$ within the XY ring. Since $n+2$ is then even, its coefficient vanishes. $\square$

Corollary 20 (Four-site ring) *For the spin-1/2 XY ring $A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$,*

$$\mathfrak{M}_\beta(AB, BC) = \frac{\beta^5 h_B}{128} \text{Im}(t_{AB} t_{BC} t_{CD} t_{DA}) + O(\beta^7). \quad (116)$$

Equation (116) agrees with the exact 16×16 certificate summarized in Appendix C. The improved remainder is restricted to this qubit XY specialization. The general unicyclic theorem has remainder $O(\beta^{n+2})$.

7.1 Exact Jordan–Wigner reconstruction at arbitrary temperature

The XY ring also admits a nonperturbative reduction. We use the Jordan–Wigner fermionization [18], with the periodic-chain parity decomposition familiar from the XY-chain solution of Lieb, Schultz, and Mattis [19]. Let $c = (c_0, \dots, c_{n-1})^T$ be the Jordan–Wigner fermions and let $p = \pm 1$ label fermion parity. In this convention, let E_p be the $n \times n$ one-particle matrix with diagonal entries $-h_j$, hopping $\bar{t}_j/2$ on the bulk links, and boundary hopping $-p \bar{t}_{n-1}/2$. Define

$$M_{p,s} = \beta E_p - \text{si} \pi I, \quad G(M) = e^{-c^\dagger M c}, \quad s = 0, 1, \quad (117)$$

and, for a contiguous block $\mathcal{B}$,

$$T_{p,s} = \text{Tr } G(M_{p,s}), \quad \Gamma_{\mathcal{B}}(M) = \text{Tr}_{\overline{\mathcal{B}}} G(M), \quad D = \sum_{p=\pm 1} \sum_{s=0,1} p^s T_{p,s}. \quad (118)$$

The unnormalized formulation is regular even where a parity-inserted Gaussian trace vanishes. Let $F_m = (\sigma^x)^{\otimes m}$ denote the fixed conversion from the Fock occupation convention to the product spin basis used in Eq. (113).

Proposition 21 (Fixed-block reconstruction) *In the spin basis used above, the reduced state on any contiguous block of m sites is*

$$\rho_{\mathcal{B}} = F_m \frac{\sum_{p=\pm 1} \sum_{s=0,1} p^s \Gamma_{\mathcal{B}}(M_{p,s})}{D} F_m. \quad (119)$$

Consequently, $\mathfrak{M}_\beta(AB, BC)$ at arbitrary n and β is obtained from four $n \times n$ Gaussian matrices and reduced matrices of size at most 8×8 . The cost grows as $O(n^3)$ rather than $O(2^{3n})$.

The proof, parity conventions, and the corresponding closed formula for $\mathfrak{M}_\beta$ are given in Appendix D. Direct comparison with exact diagonalization for $3 \leq n \leq 7$ gives agreement at machine precision for complex couplings and generic fields. High-precision evaluation at $t = (1, \dots, 1, i)$ and $h_B = 1$ through $n = 12$ reproduces the coefficient in Corollary 19 to relative discrepancy below 1.3×10^{-19} . These calculations check the XY specialization by a pipeline independent of the high-temperature coefficient extraction.

This construction supplies an exact finite-temperature realization of the free-fermion direction identified among the open problems of Ref. [2].

7.2 Thermal attenuation of the winding response

Write $W_n = (\prod_j |t_j|) e^{i\vartheta_n}$, where $\vartheta_n = \arg W_n$. The leading term in Corollary 19 can then be written as

$$\mathfrak{M}_\beta^{\text{lead}}(AB, BC) = 2(-1)^n (\beta h_B) \sin \vartheta_n \prod_{j=0}^{n-1} \frac{\beta |t_j|}{4}. \quad (120)$$

Thus the high-temperature winding process is attenuated by one factor $\beta |t_j|/4$ for every bond added outside the fixed three-site window. The parity-inserted terms in Proposition 21 identify the finite-temperature continuation of this scale.

Lemma 22 (Parity-insertion determinant) *For both fermion-parity sectors and every β ,*

$$\frac{T_{p,1}}{T_{p,0}} = \det \tanh \left(\frac{\beta E_p}{2} \right). \quad (121)$$

Proof. If $\lambda_1, \dots, \lambda_n$ are the eigenvalues of E_p , then $e^{-(\beta \lambda_j - i\pi)} = -e^{-\beta \lambda_j}$. The ratio of the two Gaussian traces is therefore $\prod_j (1 - e^{-\beta \lambda_j}) / (1 + e^{-\beta \lambda_j}) = \prod_j \tanh(\beta \lambda_j / 2)$. $\square$

For a zero-field homogeneous bulk with $J_\perp > 0$, put $a = \beta J_\perp$ and define the exact fermion-parity-string attenuation factor and its associated length by

$$r_P(a) = \exp \left\{ \frac{1}{2\pi} \int_{-\pi}^{\pi} \log \left| \tanh \left(\frac{a}{2} \cos k \right) \right| dk \right\}, \quad (122)$$

$$\xi_P(a) = \frac{1}{-\log r_P(a)}. \quad (123)$$

The dispersion entering Eq. (122) is $\varepsilon(k) = J_\perp \cos k$ in the conventions of Eq. (113). Equation (121) makes r_P the bulk geometric mean of the parity-insertion eigenvalues, or equivalently the exponential scale of the thermal fermion-parity string. The derivation in Appendix D gives

$$r_P(a) = \frac{a}{4} \left[1 - \frac{a^2}{24} + O(a^4) \right], \quad a \longrightarrow 0, \quad (124)$$

$$\xi_P(a) = \frac{2a}{\pi} [1 + O(a^{-2})], \quad a \longrightarrow \infty. \quad (125)$$

The first limit recovers the factor in Eq. (120). The second gives a parity-string thermal length proportional to inverse temperature.

We next test whether this exact parity scale also controls the modular response. Consider the one-parameter family

$$t_0 = \dots = t_{n-2} = J_\perp, \quad t_{n-1} = iJ_\perp, \quad h_1 = J_\perp, \quad h_j = 0 \quad (j \neq 1), \quad (126)$$

for which $W_n = iJ_\perp^n$, and write $\mathfrak{M}_{a,n}$ for the resulting response. Figure 1 compares the consecutive-size ratios

$$R_n(a) = \left| \frac{\mathfrak{M}_{a,n}}{\mathfrak{M}_{a,n-1}} \right| \quad (127)$$

with $r_P(a)$. Calculations at six values from $a = 0.5$ to $a = 16$, using sizes up to $n = 70$, show the ratios approaching the value in Eq. (122) over the computed sizes. The full numerical table is reported in Appendix D. No large- n value is used to determine r_P .

This agreement is a numerical identification, not an analytic consequence of Lemma 22. The latter proves the scale carried by the parity-inserted Gaussian traces, while a proof that the nonlinear modular functional has the same asymptotic rate would additionally require control of its local response and of the remaining finite-size terms. The data are consistent with the perturbative attenuation in Eq. (120) being the high-temperature limit of a geometric suppression throughout the temperature range studied.

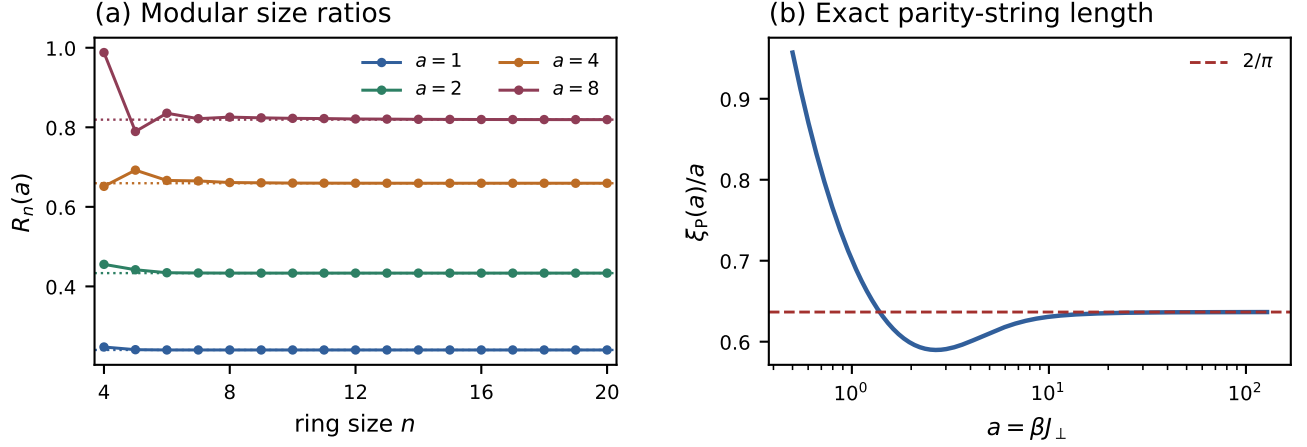


Figure 1: Thermal attenuation with ring length for the homogeneous-hopping family in Eq. (126). (a) Consecutive-size ratios $R_n(a)$ from the exact Jordan–Wigner reconstruction. Horizontal lines are the unfitted values $r_P(a)$ from Eq. (122). (b) Exact parity-string length $\xi_P(a)/a$ obtained from the integral. The dashed line is the low-temperature limit $2/\pi$. The modular data are evaluated only through $n = 20$ in the figure. The longer-size checks are given in Table 2.

8 Comparison with two spin-chirality diagnostics

We now restrict the comparison to the axial spin-1/2 triangle. The modular response is best interpreted by comparing it with two quantities on the same thermal state. The first is the scalar spin chirality [20],

$$\chi_{ABC} = \mathbf{S}_A \cdot (\mathbf{S}_B \times \mathbf{S}_C). \quad (128)$$

The second is the oriented loop of bond coherences

$$G_\beta = \text{Im} [\langle S_A^+ S_B^- \rangle_\beta \langle S_B^+ S_C^- \rangle_\beta \langle S_C^+ S_A^- \rangle_\beta]. \quad (129)$$

Unlike (1), G_β presupposes the three bond operators and the cycle whose flux is to be tested.

Direct expansion of the Gibbs state gives, on every oriented edge,

$$\langle S_x^+ S_y^- \rangle_\beta = -\frac{\beta}{8} \overline{t_{xy}} + O(\beta^2), \quad (130)$$

and hence

$$G_\beta = \frac{\beta^3}{512} \Phi_\Delta + O(\beta^4). \quad (131)$$

The leading term is gauge invariant, independent of all fields and all longitudinal exchanges, and vanishes when the cycle is opened into a tree.

By contrast,

$$\langle \chi_{ABC} \rangle_\beta = -\frac{\beta^2}{32} (h_A D_{BC}^z + h_B D_{CA}^z + h_C D_{AB}^z) + O(\beta^3). \quad (132)$$

This is a sum of single-bond DM amplitudes. The fixed laboratory operator (128) is not invariant under independent axial rotations of the three local frames, and its leading thermal expectation can remain nonzero on the open chain $A - B - C$, where the bond phases are removable and (1) vanishes exactly.

The comparison is summarized in Table 1. On the narrow task of measuring the flux of a chosen triangle, G_β has a lower-order onset in β and does not require a magnetic field. Among the three diagnostics compared here, only the modular commutator is specified by the density matrix and the

tripartition alone. More generally, its local-unitary invariance and conjugation oddness imply that a nonzero value witnesses state chirality in the sense of Ref. [13]. It does so without choosing a microscopic loop operator. The witness is not faithful. A zero modular commutator need not imply a local real structure or the absence of broader many-body chirality [15].

9 Finite-temperature crossover

To connect the perturbative coefficients with the full thermal response, we diagonalize two XY Hamiltonians. The first is the minimal triangle

$$t_{AB} = t_{BC} = 1, \quad t_{CA} = i, \quad h_A = h_C = 0, \quad h_B = 0.7, \quad (133)$$

and the second is the four-site ring

$$(t_{AB}, t_{BC}, t_{CD}, t_{DA}) = (1, 1, 1, i), \quad (h_A, h_B, h_C, h_D) = (0, 0.7, 0, 0). \quad (134)$$

All J_e^z vanish, and energies are measured in units of $|t_{AB}|$. The reduced density matrices are formed from the exact Gibbs states and their logarithms are evaluated spectrally.

Figure 2(a) displays both full responses. The triangle signal grows in magnitude from its quartic law, reaches a broad minimum near $\beta = 5.81$, and then returns to zero. Panels (b) and (c) test the triangle and square high-temperature coefficients without fitting. For the triangle,

$$m_4 = -\frac{0.7}{32}, \quad m_6 = \frac{4(0.7)^3 + 8(0.7) + 6(0.7)}{1536}, \quad (135)$$

while the square coefficient is

$$m_5^\square = \frac{0.7}{128}. \quad (136)$$

The sextic triangle approximation tracks the exact ratio over a visibly larger interval than the quartic constant alone. At the point (133), the XY quintic term vanishes and $\text{Im}(W_\Delta^2) = 0$ also removes the septic term. Thus the residual after $m_4 + \beta^2 m_6$ starts at order β^4 in the plotted ratio, providing a simultaneous numerical check of both cancellations. For the even four-site ring, the parity relation $\mathfrak{M}_{-\beta} = -\mathfrak{M}_\beta$ implies $\mathfrak{M}_\beta/\beta^5 = m_5^\square + O(\beta^2)$, as observed in panel (c).

The opposite low-temperature limits have a state-theoretic explanation. If ABC is the whole system and $\rho = |\psi\rangle\langle\psi|$ is pure, then

$$\mathfrak{M}_\rho(AB, BC) = 0, \quad (137)$$

where the logarithms are restricted to the supports of the reduced states [3].¹ Consequently, for the triangle with a nondegenerate ground state, $\mathfrak{M}_\beta \rightarrow 0$ as $\beta \rightarrow \infty$. Although the matrix logarithm is singular at the rank-deficient boundary, the divergent reduced eigenvalue logarithms grow at most linearly in β , whereas excited-state weights are exponentially small in the finite spectral gap.

¹Schmidt decomposition reduces the expectation to $\langle\psi|[\log \rho_C, \log \rho_A]|\psi\rangle = 0$.

Table 1: Leading high-temperature behavior on the axial spin-1/2 triangle.

Diagnostic	Order	Gauge invariant	Leading dependence
$\langle\chi_{ABC}\rangle_\beta$	β^2	no	$\sum_x h_x D_{e \not\sim x}^z$
G_β	β^3	yes	Φ_Δ
$\mathfrak{M}_\beta(AB, BC)$	β^4	yes	$h_B \Phi_\Delta$

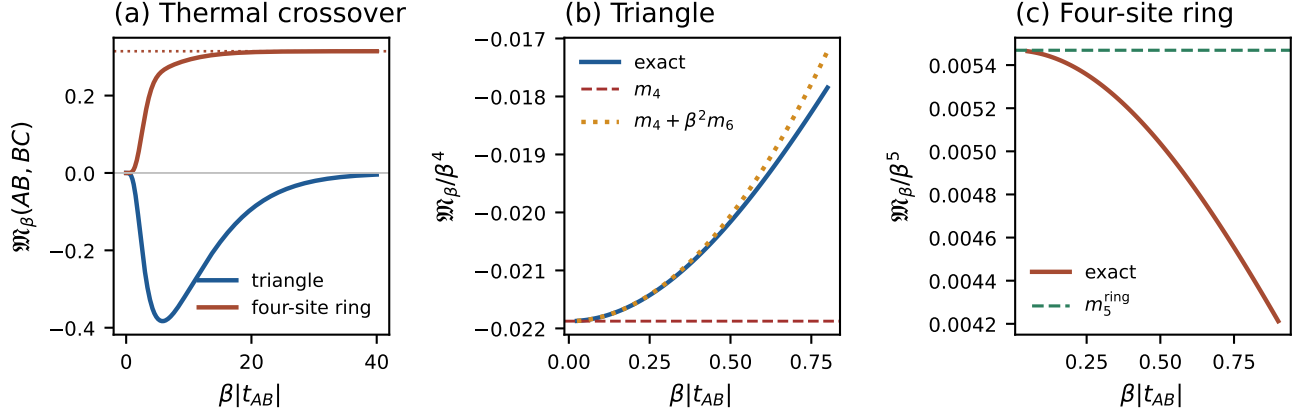


Figure 2: Finite-temperature modular response for Eqs. (133) and (134). (a) Exact-diagonalization curves for the triangle and four-site ring. (b) Triangle ratio $\mathfrak{M}_\beta/\beta^4$ compared with m_4 and $m_4 + \beta^2 m_6$. The residual is $O(\beta^4)$ because $m_5 = m_7 = 0$ at this point. (c) Four-site ratio $\mathfrak{M}_\beta/\beta^5$ compared with the exact coefficient $m_5^{\square}$. The leading correction is $O(\beta^2)$. No fitted coefficient is used.

For the four-site ring, the union $AB \cup BC = ABC$ leaves the site D outside the measured regions. The global ground state is pure and nondegenerate, but its reduced state on ABC is mixed, so Eq. (137) does not apply. At the point (134), the gap is 0.2344774934 and

$$\lim_{\beta \rightarrow \infty} \mathfrak{M}_\beta(AB, BC) \simeq 0.3146212191. \quad (138)$$

This value is obtained by direct numerical evaluation of the nondegenerate ground state. The exterior spin leaves a mixed three-site ground-state reduction, allowing a nonzero zero-temperature limit while the high-temperature loop hierarchy is preserved.

10 Measurement and scope

The results above deliberately assign a non-topological role to the modular commutator. Recent counterexamples show that it is not a universal extractor of chiral central charge on arbitrary lattices [11], while mixed-state topological phases require diagnostics beyond naive extensions of pure-state probes [12]. Our thermal statements make no such claims. They give finite-dimensional identities in specified spin models. Within the axial class, the modular commutator is an exactly controlled joint probe of a nonreal Wilson product and time-reversal breaking. It is also a nonfaithful witness of state chirality in the local-unitary sense of Ref. [13]. This is compatible with the distinct finite-depth notion of Ref. [15], which can remain nontrivial when tripartite modular commutators vanish.

For three qubits, the quantity can in principle be reconstructed from state tomography. One estimates ρ_{ABC} , obtains ρ_{AB} and ρ_{BC} by partial trace, forms their matrix logarithms, and evaluates Eq. (1). This state-intrinsic character is also the source of its principal practical cost relative to G_β . Matrix logarithms of estimated reduced states become poorly conditioned when small eigenvalues approach zero. More precisely, the Fréchet derivative obeys the scale

$$\|D \log_\rho[\delta\rho]\| \leq \frac{\|\delta\rho\|}{\lambda_{\min}(\rho)}, \quad (139)$$

so low-temperature reconstruction requires precision that grows as $\lambda_{\min}(\rho)$ decreases. Protocols for entanglement-Hamiltonian tomography have been demonstrated in quantum simulators [21], while scalar

chirality admits dedicated measurement circuits [22]. Axial DM spin models can also be engineered in programmable platforms. A Rydberg-atom construction using Raman transitions and a rotating spin frame was proposed in Ref. [23]. A full statistical error analysis for Eq. (1) lies beyond the present work.

Theorem 5 assumes a finite, simple, connected, unicyclic graph and the axial Hamiltonian (8). The spins, longitudinal exchanges, longitudinal fields, and grafted trees are arbitrary. The parameters are fixed as $\beta \rightarrow 0$. If its displayed prefactor vanishes, the theorem identifies a zero leading admissible coefficient and does not locate the first nonzero term. Proposition 12 is a coefficient extraction at one primitive multidegree. It is not an all-orders decomposition by cycles. Corollary 13 concerns only the first admissible order, even when shortest cycles overlap. The exact higher-order counterexamples in Sec. 4 preclude a general cactus law or unrestricted cycle additivity.

The axial assumption is substantive. Section 5 exhibits an open-link gauge invariant generated by transverse on-site fields. It can occur before the primitive winding term. The arbitrary-spin unicyclic coefficient itself is unchanged when extracted as the coefficient of $h_B W_\gamma$, since that extraction sets the transverse-field variables to zero. It need not dominate the complete response once those fields are present. We do not give a complete transverse-field classification or an arbitrary-spin formula for the local background.

The XY ring has a direct locality interpretation. Its fixed three-site reduced state acquires the primitive cycle phase through a process that winds around the complementary arc. At high temperature the cost of each added bond is explicit in Eq. (120). At arbitrary temperature, Lemma 22 identifies the exact scale of the fermion-parity string. The modular size ratios approach that scale over the computed sizes. We do not prove $R_n \rightarrow r_P$. We also do not extend the arbitrary-temperature attenuation analysis to XXZ rings.

11 Conclusion

For every finite unicyclic axial network considered here, the first admissible cycle coefficient is fixed by the cycle length, the field at the overlap, the product of local spin moments, and the imaginary part of the Wilson product. The result permits arbitrary irreducible spins, longitudinal exchanges, longitudinal fields, grafted trees, and nonconsecutive marked sites. The same coefficient extraction isolates primitive sectors on general graphs and the sum over shortest marked cycles at the first admissible order. Exact higher-order counterexamples show why these statements do not extend to an all-orders cycle decomposition.

The local-real-structure and antiunitary vanishing principles are state-level statements. Corollaries 3 and 4 translate them into the graph and parameter language of the axial Hamiltonian class. The winding theorem also requires that class. Transverse on-site fields generate a local open-link gauge invariant at order seven. In the seven-cycle example it appears before the primitive cycle term. A nonzero modular commutator outside the axial class therefore need not identify a closed-loop phase. The axial hypothesis is consequently a substantive boundary of the onset theorem.

For the qubit XY ring, Jordan–Wigner fermionization reduces the full finite-temperature calculation to one-particle matrices and fixed local blocks. The parity determinant and the asymptotics of ξ_P are analytic. The approach of the modular size ratios to r_P over the computed sizes is numerical. No large-size limit is claimed. The XY-ring calculation gives a finite-temperature instance relevant to the free-fermion direction in Open Problem 3 of Ref. [2]. The finite spin networks give instances relevant to the few-body question in Open Problem 7.

Scalar chirality, a coherence Wilson loop, and the modular commutator probe different information in the same axial qubit triangle. The modular quantity is less direct as a flux meter than the coherence loop. It is also the only one defined by the state and the tripartition alone. Under the hypotheses proved here, this state-intrinsic quantity has an exact local, thermal, and geometric interpretation whose limitations are explicit.

Data Availability

The source code and exact-arithmetic certificates supporting this study are available in the accompanying GitHub repository at <https://github.com/JeromeNicolas2026/thermal-modular-commutators>. The repository version used for this article is archived on Zenodo under DOI 10.5281/zenodo.22860690. The archive includes `SHA256SUMS_v06.txt`, which records SHA-256 checksums for the deposited files. No additional data were generated.

A Exact Pauli certificate for the triangle

At the parameter point (51), introduce

$$a = H_{AB} = S_A^x S_B^x + S_A^y S_B^y, \quad (140)$$

$$b = H_{BC} = S_B^x S_C^x + S_B^y S_C^y, \quad (141)$$

$$c = H_{CA} = S_A^y S_C^x - S_A^x S_C^y, \quad (142)$$

$$H_0 = a + b + c. \quad (143)$$

The normalized trace is $\tau = \text{Tr}/8$. At zero field,

$$P_0 = a, \quad Q_0 = b, \quad (144)$$

$$A_2^{(0)} = \frac{1}{8}I + \frac{1}{4}(S_A^y S_B^x - S_A^x S_B^y), \quad (145)$$

$$B_2^{(0)} = \frac{1}{8}I + \frac{1}{4}(S_B^y S_C^x - S_B^x S_C^y). \quad (146)$$

Put

$$H(h) = H_0 + hH_1^{(r)}, \quad H_1^{(r)} = -S_r^z, \quad r \in \{A, B, C\}. \quad (147)$$

The first derivatives of $P = \mathbb{E}_{AB}H$ and $Q = \mathbb{E}_{BC}H$ are

r	$P_1^{(r)}$	$Q_1^{(r)}$
A	$-S_A^z$	0
B	$-S_B^z$	$-S_B^z$
C	0	$-S_C^z$

(148)

Since $\{S^z, S^x\} = \{S^z, S^y\} = 0$ for spin 1/2, direct substitution into Eqs. (38) and (39) gives

$$\left. \frac{dA_2}{dh} \right|_{h=0} = \left. \frac{dB_2}{dh} \right|_{h=0} = 0 \quad (149)$$

for all three choices of r .

Write the three terms on the right-hand side of Eq. (43) as

$$T_0 = \frac{i}{2}\tau(H^2[P, Q]), \quad (150)$$

$$T_1 = i\tau(H[P, B_2]), \quad (151)$$

$$T_2 = i\tau(H[A_2, Q]). \quad (152)$$

Their field derivatives at zero are

$$\left. \frac{dT_0}{dh} \right|_0 = \frac{i}{2}\tau \left(\{H_0, H_1^{(r)}\}[P_0, Q_0] + H_0^2([P_1^{(r)}, Q_0] + [P_0, Q_1^{(r)}]) \right), \quad (153)$$

$$\left. \frac{dT_1}{dh} \right|_0 = i\tau \left(H_1^{(r)}[P_0, B_2^{(0)}] + H_0[P_1^{(r)}, B_2^{(0)}] \right), \quad (154)$$

$$\left. \frac{dT_2}{dh} \right|_0 = i\tau \left(H_1^{(r)}[A_2^{(0)}, Q_0] + H_0[A_2^{(0)}, Q_1^{(r)}] \right). \quad (155)$$

For $r = B$, the required Pauli identities are

$$[a, b] = iS_B^z(S_A^x S_C^y - S_A^y S_C^x) = -iS_B^z c, \quad (156)$$

$$\{a, c\} = \frac{1}{2}(S_B^y S_C^x - S_B^x S_C^y), \quad (157)$$

$$\{b, c\} = \frac{1}{2}(S_A^y S_B^x - S_A^x S_B^y). \quad (158)$$

Moreover,

$$C_1^{(B)} = [-S_B^z, b] + [a, -S_B^z] = i(S_B^x S_C^y - S_B^y S_C^x + S_A^x S_B^y - S_A^y S_B^x). \quad (159)$$

Because $\{a, S_B^z\} = \{b, S_B^z\} = 0$ and $\{c, S_B^z\} = 2cS_B^z$, the first part of Eq. (153) is

$$\frac{i}{2}\tau(-\{H_0, S_B^z\}[a, b]) = -\tau(c^2(S_B^z)^2) = -\frac{1}{32}. \quad (160)$$

Only the anticommutators (157) and (158) contribute to $\tau(H_0^2 C_1^{(B)})$. Each orientation gives $1/32$ after multiplication by $i/2$, hence

$$\frac{i}{2}\tau(H_0^2 C_1^{(B)}) = \frac{1}{16}. \quad (161)$$

Combining Eqs. (160) and (161) yields

$$\left. \frac{dT_0}{dh} \right|_{0, r=B} = \frac{1}{32}. \quad (162)$$

Equations (145) and (146) also give

$$[-S_B^z, B_2^{(0)}] = \frac{i}{4}b, \quad [A_2^{(0)}, -S_B^z] = \frac{i}{4}a. \quad (163)$$

The remaining terms in Eqs. (154) and (155) vanish by local Pauli orthogonality. Since $\tau(a^2) = \tau(b^2) = 1/8$, one finds

$$\left. \frac{dT_1}{dh} \right|_{0, r=B} = \left. \frac{dT_2}{dh} \right|_{0, r=B} = -\frac{1}{32}. \quad (164)$$

For $r = A$, one has $Q_1^{(A)} = 0$ and $[P_1^{(A)}, Q_0] = 0$. Furthermore, $\{H_0, S_A^z\} = 2bS_A^z$, and every remaining trace contains on A either $S_A^z S_A^x$ or $S_A^z S_A^y$. It therefore vanishes. The $r = C$ calculation is the reflected counterpart. Altogether,

r	dT_0/dh	dT_1/dh	dT_2/dh	dm_4/dh
A	0	0	0	0
B	$1/32$	$-1/32$	$-1/32$	$-1/32$
C	0	0	0	0

(165)

This is the spin-1/2 specialization of Eq. (55).

B Exact identification of the fifth- through seventh-order coefficients

The higher coefficients were obtained without floating-point arithmetic. For a truncation order N , we form

$$r_\beta = \frac{\sum_{n=0}^N \frac{(-\beta)^n}{n!} H^n}{\tau \left(\sum_{n=0}^N \frac{(-\beta)^n}{n!} H^n \right)} + O(\beta^{N+1}) \quad (166)$$

and use the finite series

$$L_X(\beta) = \sum_{p=1}^N \frac{(-1)^{p+1}}{p} (\mathbb{E}_X(r_\beta) - I)^p + O(\beta^{N+1}). \quad (167)$$

Substitution into Eq. (35) reduces every coefficient to Pauli matrix products with entries in $\mathbb{Q}(i)$.

For m_5 , the classification (105) leaves nine constants. Setting $t = (1, 1, i)$ and turning on one J_e^z at a time gives, in units of $1/384$,

$$\begin{array}{c|ccc} & J_{AB}^z & J_{BC}^z & J_{CA}^z \\ \hline h_A & 3 & -2 & 0 \\ h_B & -1 & -1 & -3 \\ h_C & -2 & 3 & 0 \end{array} \quad (168)$$

which is Eq. (106). Reflection $A \leftrightarrow C$ together with $AB \leftrightarrow BC$ provides an independent covariance check.

For the XY sextic sector, write

$$\frac{m_6}{\Phi_\Delta} = p_3(h_A, h_B, h_C) + \sum_{x,e} b_{x,e} h_x |t_e|^2, \quad (169)$$

where p_3 is cubic. The cubic part is already determined at one nonzero flux point and is found to be

$$p_3 = \frac{h_B(h_A^2 + h_A h_C + h_B^2 + h_C^2)}{384}. \quad (170)$$

The nine linear-amplitude coefficients are fixed from the three configurations whose squared amplitudes form

$$\mathcal{C} = \begin{pmatrix} 1 & 1 & 1 \\ 4 & 1 & 1 \\ 1 & 1 & 4 \end{pmatrix}, \quad \det \mathcal{C} = -9. \quad (171)$$

The exact solution is

$$\begin{array}{c|ccc} & |t_{AB}|^2 & |t_{BC}|^2 & |t_{CA}|^2 \\ \hline h_A & 1/1536 & 0 & 0 \\ h_B & 1/384 & 1/384 & 1/256 \\ h_C & 0 & 1/1536 & 0 \end{array}. \quad (172)$$

Equations (170) and (172) reproduce the hopping polynomial (108). Four additional hopping configurations, not used in the identification, were checked against the closed form.

The remaining sextic sector is quadratic in longitudinal exchanges. In units of $\Phi_\Delta/1536$, its eighteen coefficients are

$$\begin{array}{c|ccc|ccc} & (J_{AB}^z)^2 & (J_{BC}^z)^2 & (J_{CA}^z)^2 & J_{AB}^z J_{BC}^z & J_{BC}^z J_{CA}^z & J_{AB}^z J_{CA}^z \\ \hline h_A & 1 & 0 & 0 & 0 & -4 & 3 \\ h_B & 1 & 1 & 2 & 1 & -1 & -1 \\ h_C & 0 & 1 & 0 & 0 & 3 & -4 \end{array} \quad (173)$$

Six configurations of $(J_{AB}^z, J_{BC}^z, J_{CA}^z)$ isolate these six quadratic monomials while retaining symbolic fields. Two additional generic choices with $t_e \in \mathbb{Q}(i)$ and $J_e^z \in \mathbb{Q}$, not used in the calibration, reproduce Eq. (109) exactly. This establishes the full sextic formula (110).

Finally, in the XY model the degree-seven classification leaves three constants:

$$m_7 = \text{Im}(W_\Delta^2)(q_A h_A + q_B h_B + q_C h_C). \quad (174)$$

Exact evaluation with symbolic fields and $t_e \in \mathbb{Q}(i)$ at a point with $\text{Im}(W_\Delta^2) = 1440$ gives

$$(q_A, q_B, q_C) = -\frac{1}{92160}(2, 41, 2). \quad (175)$$

A second generic point with $\text{Im}(W_\Delta^2) = -378$ and the unit-amplitude parameter point $(1, 1, i)$, where this invariant vanishes, provide two exact checks.

C Exact four-site coefficient certificate

The formal construction from (166) through (167) also applies to the 16×16 four-site Hamiltonian. With symbolic h_A, h_B, h_C, h_D and exact arithmetic over $\mathbb{Q}(i)$, evaluation at the XY point $(1, 1, 1, i)$ gives

$$[\beta^5]\mathfrak{M}_\beta(AB, BC) = \frac{h_B}{128}, \quad [h_A] = [h_C] = [h_D] = 0. \quad (176)$$

Replacing $t_{AB} = 1$ by $t_{AB} = 2$ doubles the result. Gauge-flow grading at degree five permits only $\text{Im } W_4 \sum_x c_x h_x$. Hence these evaluations determine all four constants and reproduce Corollary 20. Generic rational longitudinal exchanges provide an additional exact check: they cannot enter the degree-five winding monomial because its four hopping factors and one field already exhaust the degree.

D Exact fermionic reconstruction of XY rings

We give the details behind Proposition 21. Let c_j be Jordan–Wigner fermions ordered from 0 to $n-1$, let $\hat{p} = (-1)^{N_f}$, and write

$$S_j^z = c_j^\dagger c_j - \frac{1}{2}, \quad S_{n-1}^+ S_0^- = -\hat{p} c_{n-1}^\dagger c_0. \quad (177)$$

All bulk strings cancel. In the sector $\hat{p} = p$, the Hamiltonian is $Q_p + \kappa_h$, where $Q_p = c^\dagger E_p c$, $\kappa_h = \frac{1}{2} \sum_j h_j$, and E_p is the one-particle matrix specified below Eq. (117). The projectors $\Pi_p = (I + p\hat{p})/2$ give

$$e^{-\beta H_n} = \frac{e^{-\beta \kappa_h}}{2} \sum_{p=\pm 1} \left(e^{-\beta Q_p} + p\hat{p} e^{-\beta Q_p} \right), \quad (178)$$

$$\hat{p} e^{-\beta Q_p} = e^{-c^\dagger (\beta E_p - i\pi I) c}. \quad (179)$$

Thus the four matrices in Eq. (117) arise directly from the two parity sectors, with and without a parity insertion. The scalar $e^{-\beta \kappa_h}/2$ cancels on normalization.

For any matrix M , the number-conserving Gaussian operator satisfies

$$T(M) := \text{Tr } G(M) = \det(I + e^{-M}), \quad C(M) = (e^M + I)^{-1}. \quad (180)$$

When $T(M) \neq 0$, the normalized operator $G(M)/T(M)$ reduces on a contiguous set $\mathcal{B}$ to the Gaussian operator whose correlation matrix is the principal submatrix $C(M)|_{\mathcal{B}}$. Equation (118) is preferable because it remains regular at zeros of a parity-inserted trace. Indeed, $G(M_{p,1}) = \hat{p} e^{-\beta Q_p}$ is regular even though $T(M_{p,1}) = \det(I - e^{-\beta E_p})$ vanishes at $\beta = 0$.

We now derive the bulk scale used in Eqs. (122) through (125). In a zero-field homogeneous ring, the eigenvalues of E_p are $J_\perp \cos k_m$, where the parity and the total flux only shift the uniform momentum grid. Lemma 22 therefore gives

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left| \frac{T_{p,1}}{T_{p,0}} \right| = \frac{1}{2\pi} \int_{-\pi}^{\pi} \log \left| \tanh \left(\frac{a}{2} \cos k \right) \right| dk = \log r_P(a). \quad (181)$$

The logarithmic singularities at the two Fermi points are integrable. A momentum grid that meets a zero is understood by a limiting twist. Thus ξ_P is the attenuation length of the thermal fermion-parity insertion around the homogeneous ring.

For $a \rightarrow 0$, the expansion

$$\log |\tanh x| = \log |x| - \frac{x^2}{3} + \frac{7x^4}{90} + O(x^6) \quad (182)$$

Table 2: Large-size numerical check of the parity-string attenuation scale. $R_N(a)$ is computed from the Jordan–Wigner reconstruction, whereas $r_P(a)$ is evaluated independently from Eq. (122).

a	N	$R_N(a)$	$r_P(a)$
0.5	13	0.123718566605	0.123718566605
1	18	0.240210848508	0.240210848508
2	25	0.433416769933	0.433416769933
4	35	0.659386662495	0.659386662495
8	50	0.819227344995	0.819227344443
16	70	0.906193444199	0.906193409422

together with $(2\pi)^{-1} \int_{-\pi}^{\pi} \log |\cos k| dk = -\log 2$ yields

$$\log r_P(a) = \log \frac{a}{4} - \frac{a^2}{24} + \frac{7a^4}{3840} + O(a^6), \quad (183)$$

which implies Eq. (124). For $a \rightarrow \infty$, only neighborhoods of $k = \pm\pi/2$ contribute algebraically. Rescaling their local coordinate by $x = a(k \mp \pi/2)/2$ and using

$$\int_{-\infty}^{\infty} -\log |\tanh x| dx = \frac{\pi^2}{4} \quad (184)$$

gives

$$-\log r_P(a) = \frac{\pi}{2a} + O(a^{-3}), \quad (185)$$

and hence Eq. (125).

Table 2 records the numerical comparison between the exact factor $r_P(a)$ and the last consecutive-size ratio computed from the full modular functional in the family (126). The adaptive cutoff keeps the modular signal above the double-precision roundoff floor. Increasing the cutoff gives the displayed finite-size approach until that floor is reached.

For the displayed cutoffs, the relative discrepancy is below 4×10^{-8} throughout and below 10^{-12} for $a \leq 4$. The larger residuals at $a = 8$ and 16 reflect the slower finite-size stabilization.

With F_m as defined before Proposition 21, taking a partial trace of Eq. (178) proves Eq. (119). In particular, with D from Eq. (118),

$$\rho_{ABC} = F_3 \frac{\sum_{p,s} p^s \Gamma_{ABC}(M_{p,s})}{D} F_3, \quad (186)$$

$$\rho_{AB} = \text{Tr}_C \rho_{ABC}, \quad \rho_{BC} = \text{Tr}_A \rho_{ABC}. \quad (187)$$

The desired quantity is then the fixed-size expression

$$\mathfrak{M}_\beta(AB, BC) = i \text{Tr} \rho_{ABC} [\log \rho_{AB} \otimes I_C, I_A \otimes \log \rho_{BC}]. \quad (188)$$

The only matrices whose dimension grows with n are E_p and their matrix functions. Hence the $O(n^3)$ cost quoted in Proposition 21.

We finally record the parity relation used in Corollary 19. Complex conjugation in the spin basis gives

$$\mathfrak{M}_\beta(\bar{t}, h) = -\mathfrak{M}_\beta(t, h). \quad (189)$$

For even n , the on-site unitary

$$U = (\sigma^x)^{\otimes n} \prod_{j=0}^{n-1} e^{i\pi j S_j^z} \quad (190)$$

obeys $UH_n(t, h)U^\dagger = -H_n(\bar{t}, h)$. Local-unitary invariance and Eq. (189) therefore imply

$$\mathfrak{M}_{-\beta}(t, h) = -\mathfrak{M}_\beta(t, h), \quad n \text{ even.} \quad (191)$$

For odd n and $\text{Re } W_n = 0$, phases α_j can instead be chosen so that $e^{i(\alpha_j - \alpha_{j+1})} = -\bar{t}_j/t_j$ around the cycle. The product unitary $\prod_j e^{-i\alpha_j S_j^z} \sigma_j^x$ then maps H_n to $-H_n$, giving

$$\mathfrak{M}_{-\beta}(t, h) = \mathfrak{M}_\beta(t, h), \quad n \text{ odd, } \text{Re } W_n = 0. \quad (192)$$

These identities justify the parity-restricted high-precision extraction used as an independent check. At $t = (1, \dots, 1, i)$ and $h_B = 1$, calculations with 80-digit arithmetic for $3 \leq n \leq 12$ reproduce $2^{2n-1}[\beta^{n+1}]\mathfrak{M}_\beta = (-1)^n$ with maximum relative discrepancy 1.3×10^{-19} . Exact diagonalization and the fermionic reconstruction agree at machine precision for $3 \leq n \leq 7$ with generic complex couplings and fields.

E Computational validation

The analytic proofs of Theorem 5, Proposition 12, and Corollary 13 do not rely on computational interpolation. The finite-dimensional coefficient identities in Propositions 14 and 15 are obtained by exhaustive exact-arithmetic expansions of the normalized Gibbs state and reduced logarithms. Further exact certificates test the arbitrary-spin factor, nonconsecutive marked sites, tree pruning, primitive-cycle coefficients, and the stated higher-order obstructions. One implementation uses exact SymPy arithmetic [24]. A separately written standard-library implementation reproduces the formal-series calculations. It is an independent implementation of the same method, not an independent mathematical method.

Direct Gibbs-series evaluations, exact diagonalization, and the four-component fermionic reconstruction provide numerical cross-checks. High-precision calculations independently check the leading ring coefficient and the parity-restricted series. The analytic attenuation formula, both of its stated asymptotic regimes, every tabulated value, and all plotted data are also checked before output is written. No fitted coefficient is used. The repository and the versioned archive identified in the Data Availability Statement contain the complete source code, exact test points, expected outputs, dependency and execution information, and a SHA-256 manifest of the deposited file set.

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